How Data Velocity Changes Decisions:

The Role of Information Access Speed

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Abstract

Recent technological advancements have enabled more data-driven decision-making across firms. While prior literature highlights the value of using more data, there is less insight on the impact of information speed, particularly how quickly decision-makers can access it. We study how faster information access influences how decision-makers acquire information and make decisions, using data from healthcare. We examine a technology that visually displayed when test results became available, accelerating information access for 64,152 decisions by 387 physicians. Faster access allowed decision-makers to gather less but more targeted information, resolving uncertainty earlier and speeding up decisions, while ultimately improving decision outcomes by supporting more effective information processing. Our findings show that investing in data velocity can yield significant benefits through both faster and better decisions.

Keywords: Data-driven decision-making, experimentation, technology, information, speed, healthcare

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1 Introduction

Recent technological advancements have expanded opportunities to collect and leverage data across the organization, leading firms to rely more on data to inform their decisions rather than intuition alone (Brynjolfsson and McElheran, 2016, 2019; Adner et al., 2019; Agrawal et al., 2019). A large and growing literature shows that the ability to access and draw on information improves decision-making and firm performance across a wide range of contexts, from entrepreneurship and innovation to competitive strategy (Camuffo et al., 2020; Koning et al., 2022; Nagaraj, 2022; Kim, 2024; Galdon-Sanchez et al., 2024). This work has primarily focused on *whether* information is used at all and on the *quantity* of data available to decision-makers.

However, recent technological change has altered not only how much information firms possess, but also its *speed* –how quickly decision-makers can access and draw on it. Firms increasingly invest in dashboards, monitoring systems, and analytic tools that reduce delays between when information becomes available and when it is accessed and processed by decision-makers. Despite the prevalence of these investments, there is limited theoretical insight and systematic evidence on the implications of faster information access for firm decision-making and performance.

While it may seem natural to expect that faster access to information leads to faster decisions, the broader implications are far less clear. Faster access may simply compress the time spent waiting for information to arrive, resulting in the same decisions made more quickly. Alternatively, faster access may change what information decision-makers choose to acquire, thereby altering the decisions themselves and how they are made. In particular, faster access allows decision-makers to process early signals in real time and adjust subsequent information requests, enabling information acquisition to become more efficient by yielding relevant insights earlier and with less information overall. Whether such changes improve outcomes depends on whether faster access enables better-targeted information acquisition and effective processing of early signals, rather than inducing premature decisions. Importantly, these mechanisms would be different from a mechanical effect due to less information decay; rather, speed may matter because it reshapes when and how uncertainty is resolved within the decision process.

In this paper, we explore the impact of information access speed on how decision-makers acquire information and make decisions, and provide empirical evidence from a healthcare context. We show that faster information access not only increases the speed of decisions but also enables a more efficient decision-making process that leads to the acquisition of less but more targeted information. Furthermore, this increase in information access speed enhances decision quality, leading to improved performance outcomes. Taken together, our findings suggest that the speed of information access can be a source of advantage by reshaping how decisions are made, not merely how quickly they are reached. This has implications for research on experimentation in decision-making, suggesting that investing in faster information access may help decision-makers improve their theories and develop better experiments (Camuffo et al., 2023; Agrawal et al., 2024; Camuffo et al., 2020; Gans, 2023; Camuffo et al., 2026). It also provides insights on how information access speed can enable the building of competitive advantages based on speed (Eisenhardt, 1989; Baum and Wally, 2003; Hawk et al., 2013). More broadly, these findings highlight the importance of understanding not only what decisions are made but *how* they are made within organizations, especially in the age of data and artificial intelligence.

We examine the impact of a technology intervention that increased the speed of information access across 64,152 decisions made by 387 physicians serving 43,607 unique patients on *non-critical* cases in the Emergency Department (ED) of a major hospital, which are those that do not require immediate physician attention.[1] This setting provided uniquely rich and detailed administrative data on a large number of decisions, allowing us to observe the entire process of decision-making across a large number of physicians: the information they acquire (e.g., the diagnostic tests ordered), how quickly they make decisions based on that information (e.g., the length of patient stay), and the performance outcomes of those decisions for the organization (e.g., quality of care and patient

[1]Importantly, all critical cases arriving by ambulance requiring immediate physician attention (i.e., those with high mortality risk) were dispatched to a separate ward in this hospital (resuscitation room). In such cases, the relationship between speed and organizational performance may be more directly linked and in part mechanical. We do not study this ward in this paper.

satisfaction).

The technology increased the speed at which physicians could access information about the availability of diagnostic test results. Prior to its implementation, physicians ordering a laboratory test were not informed when results became available.[2] Instead, physicians had to repeatedly log into the internal system and navigate several screens to manually check test status—a process that could not be delegated and was costly in a time-constrained environment. As a result, physicians could not check frequently and experienced long lags in receiving information about the tests they ordered. In June 2022, a dashboard was installed in one of the two adult wards, visually notifying physicians when test results were ready and thereby increasing the speed of information access.[3] Importantly, the intervention did not change what information was available, the quality of that information signal (i.e., information decay), or how quickly tests were processed; it only reduced the time to acquire test result information by reducing frictions in learning when results were ready. The technology was installed in only one of the two adult wards in the hospital at a time when no other organizational changes were introduced, allowing us to use a difference-in-differences strategy to study its effects (Chan, 2016).

We find four main results. First, faster information access improved the efficiency of information acquisition: physicians ordered fewer, more targeted diagnostic tests, leading to a 25% reduction in total charges billed. Second, faster information access led to faster decisions, as measured by a substantial reduction in patients' length of stay (13% with the most saturated estimation, corresponding to approximately 75 minutes). This effect is only partially explained (48%) by reduced test ordering, indicating that faster information access accelerated decision-making beyond

[2] During the entire sample period both pre- and post-intervention, the laboratory called the physicians in both wards only in the rare cases when a lab test result fell outside the normal range, and in those cases, they also communicated the exact value of the result.

[3] For each laboratory test ordered, the system displayed its processing status and whether it fell within pre-established reference ranges. See a screenshot of the dashboard in Figure A.1. These passive visual alerts were visible to all physicians and nurses in the ward but did not include specific test results, physician names, or patient names—only the patient's national ID number. No active alerts were sent to any physician devices. Physicians still had to log into the system manually to view results. In addition to signaling the availability of results, the technology also indicated the severity of those results. While this feature could theoretically influence physicians' strategic decision-making—-such as prioritizing which patient to see first-—we believe this effect is limited. Physicians were already informed, both before and after the introduction of the technology, when a test result fell within a critical range requiring urgent attention.

the effect of ordering fewer tests. Third, decision outcomes improved rather than deteriorated: hospitalization rates declined without an increase in 30-day readmissions, and patient satisfaction increased. Fourth, we find that effects of faster information access were larger when workloads were higher and decisions involved less common cases—-precisely when the value of receiving information earlier while case details are still salient would be higher—-consistent with more effective processing of early signals to better target information acquisition. Taken together, these results indicate that faster information access did not induce premature closure or gaming of performance metrics, but instead improved the quality of care, consistent with physicians making better decisions.

Contribution to the literature We contribute to two strands of literature. First, we contribute to research on data-driven decision-making, which has demonstrated the value of information for improving managerial and organizational decisions (Brynjolfsson and McElheran, 2019; Camuffo et al., 2020; Koning et al., 2022; Nagaraj, 2022). Building on prior work that highlights that *how* organizations process information often matters more than the amount of data available (Kim, 2024; Kim et al., 2024), we show that data velocity—specifically the speed of information access—is a critical determinant of how decision-makers learn and improve strategic decisions. Our results show that by accelerating access to information, faster data environments reshape how decision-makers acquire information, leading them to request less but more targeted information and to use early signals more effectively. In doing so, our study highlights a novel mechanism through which data improves decision-making: not simply by providing more information, but by changing which information is acquired and how it is processed.

Second, our paper contributes to research on speed or time as a source of competitive advantage. Much work has emphasized the importance of speed for firm performance and competitive advantage (Stalk, 1988; Eisenhardt, 1989; Stalk Jr and Hout, 1990; Forbes, 2005; Hawk et al., 2013; Teece et al., 1997; Helfat et al., 2009), through a first-mover advantage (Makadok, 1998), a series of temporary advantages (Garud et al., 1998), or improved learning via more interactions with the environment (Eisenhardt, 1989; Mosakowski, 1997). This literature has generally conceptualized speed as an advantage in competition, for example by enabling firms to more rapidly respond to customer demand and realize revenues from investments relative to competitors (Stalk, 1988; Hawk et al., 2013; Pacheco-de Almeida et al., 2015).[4] Much of this literature has also been theoretical, qualitative, or correlational—providing valuable insights but making it difficult to disentangle the impact of speed from other related changes such as flexibility (e.g., just-in-time production) or better management practices. In this paper, we study this phenomenon across thousands of decisions within an organization and highlight a novel lever through which organizations can increase decision speed—by improving the speed of information access—and highlight that beyond faster competitive moves, information speed can result in better decisions.

2 Theoretical framework

In this section, we examine how faster information access shapes firm decisions. We define information access speed and argue that, alongside information quantity, the speed at which information is accessed and used by decision-makers is a key determinant of firm decisions, even in the absence of information decay.[5] We then develop hypotheses showing how faster access reshapes information acquisition and enables faster and higher-quality decisions.

2.1 The role of information access speed on firm decisions

Information—on competitors, customers, suppliers, and internal operations—is a central input to firm decision-making. A large literature, spanning research on evidence-based management (Pfeffer and Sutton, 2006; Barends and Rousseau, 2018) and more recent work on data-driven decision-making (Brynjolfsson and McElheran, 2019), shows that the acquisition and use of information is an important driver of firm decisions and performance differences across contexts such as competitive strategy (Kim, 2024), entrepreneurship (Camuffo et al., 2020), and innovation (Nagaraj, 2022).

[4] A key exception is Eisenhardt (1989), which focuses on how speed affects decisions.

[5] Specifically, we are interested in cases where, while information might depreciate over time, the timeframe for the decay is longer than the change in information access speed—such that the effects of speed are not simply mechanical in accelerating and improving decisions.

In strategic settings characterized by uncertainty, information helps decision-makers evaluate alternatives, update beliefs, and make more informed choices (Agrawal et al., 2019; Gans et al., 2019, 2020; Agrawal et al., 2024). A large body of work in strategy and entrepreneurship further emphasizes that organizations often take actions specifically to generate information that informs future decisions (Ries, 2011; Eisenmann et al., 2012; Blank, 2013; Kerr et al., 2014; Gans et al., 2019; Camuffo et al., 2020; Leatherbee and Katila, 2020; Koning et al., 2022). For example, real options approaches highlight how initial investments generate signals that shape subsequent strategic choices (Adner and Levinthal, 2004), and a Mendelian view of how executives operate emphasizes intentionality in generating options, testing them through "experiments"—that is, actions undertaken to generate information (Murray and Tripsas, 2004; Kim, 2025)—and ultimately selecting among alternatives (Levinthal, 2017).

Despite these advances, much of this work focuses on the value of acquiring *more* information, often treating information acquisition as a largely binary or static decision (see Gans et al. (2020); Gans (2023) for exceptions). Far less attention has been paid to how decision-makers acquire information over time. This is striking, given that discussions of "big data" in management emphasize three defining attributes: the volume, velocity, and variety of available data (McAfee et al., 2012). While prior research has extensively examined how data volume and variety affect firm decisions, the implications of information velocity remain underexplored.

Conceptually, data velocity comprises three components: the speed of acquiring data, the speed of transmitting data, and the speed of processing data into usable insights (Chen et al., 2012; Bharadwaj et al., 2013; Davenport and Patil, 2012). Organizations can invest in faster data acquisition by generating or capturing data more quickly, such as through sensors in retail environments that capture customer behavior in real time (McAfee et al., 2012). They can reduce transmission delays through investments in infrastructure that lower latency, as financial firms do by investing in high-speed connections for market data (Brogaard et al., 2014). Finally, organizations can accelerate data processing by increasing the speed at which insights are generated and made accessible—either by enabling faster human attention to information or by automating the transformation of raw data into insights. Examples include edge computing technologies that analyze streaming data in near real time to automatically feed into decisions, as well as various visual, alert, and monitoring technologies that ensure faster delivery of insights to decision-makers (Shi et al., 2016). A common way in which firms invest in the speed of information access is through dashboards that display information in real-time, adopted by firms especially across tech, retail, and manufacturing (Berman and Israeli, 2022; Stein et al., 2025; Team, 2022)—which enable decision-makers to quickly access and attend to information without searching through or logging into access reports.

Among these elements, the speed of information access—how quickly decision-makers can observe and use available information—ultimately determines whether data translate into strategic advantage. Organizations may excel at collecting or transmitting information, yet fail strategically if decision-makers cannot access and leverage it when needed (Kim, 2024). Real-time satellite data, for instance, offers little strategic value if executives cannot incorporate it into their decisions with comparable speed. This bottleneck—the time between information availability and use—shapes whether organizations capitalize on their information capabilities and generate temporal advantages, even when competitors eventually obtain the same information.

Accordingly, we focus on information access speed, defined as the time between when information becomes available and when decision-makers can observe and use it. Increasing information access speed does not alter the information set itself. Rather, it reduces frictions in detecting the availability of existing information—for example, by making newly available results visible sooner through dashboards or monitoring tools. By changing when information enters the decision process, faster access can affect decisions even when the content, interpretation, and validity of the information remain unchanged. Importantly, our argument does not rely on information decay: it applies in settings where any potential decay occurs on a longer timescale than the decision itself and the changes in information access speed we study.

2.2 How information access speed impacts decisions

While accessing information more quickly could lead to the same decisions simply being made faster, we argue that increased speed may instead change—and improve—how decisions are made by altering what information decision-makers acquire and how they process it.

First, faster information access can enable the identification of more relevant and targeted information, even when holding the informativeness of the information constant, allowing decision-makers to ultimately require less information overall. When early results become available quickly, decision-makers can refine subsequent information requests based on what they have already learned, rather than continuing to acquire information broadly or in parallel. This concentrates effort on the most promising lines of inquiry, reducing unnecessary information gathering. Research on sequential and Bayesian experimentation highlights how focusing information acquisition on the most informative sources can improve efficiency while preserving learning value (e.g., Berry, 2004; Mao and Bojinov, 2021; Bojinov and Gupta, 2022).

For example, in product development, companies often conduct multiple rounds of user interviews, usability studies, and pilot deployments in parallel to investigate how to increase growth. Faster access to early usage and engagement data—such as where users abandon a workflow or repeatedly encounter errors—allows decision-makers to identify which aspects of a feature warrant deeper investigation. Subsequent information gathering can then focus on those dimensions, rather than continuing broad and unfocused testing. In this way, faster access shifts information acquisition toward fewer, more targeted tests, before a final decision is reached. This leads us to our first hypothesis:

Hypothesis 1. *Increased speed in information access results in less, more targeted information acquisition for decisions.*

Second, beyond the mechanical effects of faster access itself or fewer information requests, the effect of information access speed on decision time is theoretically ambiguous. Two mechanical channels are straightforward. First, faster access reduces waiting time between when information becomes available and when it can be used. Second, if faster access leads to fewer information requests, decision-makers may spend less time waiting simply because there is less information to wait for.

Beyond these effects, however, faster information access may also affect decision time through how uncertainty is resolved. Prior research shows that the availability of additional information—particularly information that arrives early or is easily observed—does not necessarily improve inference and can instead distort attention or slow convergence. Decision-makers may overweight early but weakly informative signals (the "streetlight effect") or engage in excessive rechecking and verification, increasing cognitive complexity and delaying action (e.g., Deck and Jahedi, 2015; Hoelzemann et al., 2024). As a result, although faster access can mechanically reduce waiting time, it does not automatically translate into faster decisions.

We argue that when faster information access improves the targeting of information acquisition (i.e., Hypothesis 1), it can reduce decision time by enabling earlier resolution of uncertainty. Many firm decisions rely on implicit stopping rules: decision-makers proceed once uncertainty has been reduced below a threshold sufficient to act. When information is less targeted, decision-makers may continue deliberating simply because they have not yet observed sufficiently informative signals that cross this threshold. If faster access allows more informative signals to be observed earlier, it can resolve uncertainty sooner and accelerate the decision.

For example, consider a firm deciding whether to launch a new product nationally after a pilot. Managers monitor multiple indicators—such as customer reviews and regulatory feedback—but evidence of a sustained safety or compliance issue alone may be sufficient to halt the launch. Faster access to early quality or compliance data allows managers to recognize sooner when uncertainty has been resolved and a go/no-go threshold has been crossed, accelerating the decision. Similarly, although managers may track multiple performance indicators in market exit decisions, evidence that a unit falls below a viability threshold can be sufficient to prompt exit. Faster access to margin and demand data allows managers to recognize sooner when this condition is met. In such cases, faster access accelerates the decision not by decreasing the amount of information acquired, but because decisive information becomes visible earlier, allowing uncertainty to be resolved sooner. This leads us to our second hypothesis:

Hypothesis 2. *Increased speed in information access leads to faster decisions beyond mechanical reductions in waiting time or information acquisition, reducing the time required to reach a decision.*

Faster information access can not only change what information decision-makers acquire and how quickly uncertainty is resolved, but also how that information is processed. Importantly, neither more targeted information acquisition nor faster resolution of uncertainty necessarily implies better decisions. A natural concern is that acquiring less information and deciding more quickly could reflect premature closure—acting on insufficient information or truncating deliberation too early.

Whether decision quality improves therefore depends on how changes in the timing of informative signals affect the processing of information. Research in cognitive science and behavioral decision-making shows that attention and working memory are limited (e.g., Miller, 1956; Cowan, 2001), that interruptions and task switching impair the integration of information (e.g., Pashler, 1994; Leroy, 2009), and that cognitive load reduces inferential accuracy (e.g., Mani et al., 2013; Simon, 1991). Information is more effectively interpreted when it arrives while the decision-maker remains cognitively engaged with the problem—when relevant details, hypotheses, and alternatives are still salient—rather than after attention has shifted to other tasks or decisions.

By reducing delays between information availability and use, faster information access can align the arrival of diagnostic signals with active consideration. In doing so, it can improve inference and belief updating—not because the information itself is mechanically better or more informative, but because it is processed at a moment of active consideration when relevant details and hypotheses remain salient. This leads us to our final hypothesis:

Hypothesis 3. *Increased speed in information access leads to improvements in the quality of decisions, even when the informational content of signals does not decay over the decision time frame.*

In particular, these improvements should be more pronounced in settings where earlier access to information reduces cognitive frictions—such as task switching or cognitive load—making relevant information easier to interpret for the decision.

Taken together, our argument is that organizations can actively shape when and how uncertainty is resolved by investing in information access speed. By altering how quickly relevant information becomes visible to decision-makers, organizations can influence which information is acquired, when stopping thresholds are reached, and how information is processed. In this way, information access speed functions as an organizational design lever that governs the timing of learning and decision-making, rather than merely accelerating access to information that would otherwise be used in the same way.

In the following sections, we empirically examine whether increasing the speed of information access reshapes the information decision-makers choose to acquire, and whether these changes enable faster and higher-quality decisions in a healthcare context.

3 Empirical context

We first outline the empirical requirements for studying this question and how our setting—a hospital emergency department—meets them. We then introduce the details of our setting and the technology intervention that increased the speed of information access.

3.1 Empirical setting

Studying the effects of information access speed on strategic decision-making imposes several empirical challenges. First, it requires access to a large number of complex, high-stakes decisions made by a diverse set of decision-makers. Because decision-making often occurs under uncertainty, a large sample of a common set of decisions is needed to evaluate whether decisions improve performance *in expectation*, as individual outcomes may not reveal underlying decision quality. Second, we must observe the full decision-making process, including what information is acquired, how it is used, how quickly decisions are made, and their performance consequences. This level of granularity is difficult to capture in most settings, as high-stakes decisions are rarely timestamped, documented in detail, or consistently tracked. Third, we need a substantial and plausibly exogenous change in the speed of information access to credibly isolate its effect from other factors, such as improved management practices or increased data availability.

To meet these requirements, we study a natural experiment in the Emergency Department of Fundación Valle del Lili (FVL), a leading teaching hospital in Cali, Colombia. Ranked 126th globally in Newsweek's 2026 list of the World's Best Hospitals (Newsweek, 2026), FVL operated 680 beds and 20 operating rooms in 2021, staffed by 724 physicians and serving nearly half a million outpatient visits and 36,000 hospital discharges annually. What makes this setting distinctive is that it provides real-time data on decisions made by hundreds of physicians, under high stakes, with detailed timestamps and measurable outcomes —features rarely available together. This allows us to track how decision-making changes when the speed of information access varies.

Our setting builds on a growing tradition in the strategic management literature that uses hospitals to study strategic decision-making. Prior work has leveraged healthcare settings to examine how organizations navigate uncertainty, allocate resources, and manage trade-offs between quality and efficiency. For example, Douglas and Ryman (2003) studied competitive advantage in U.S. hospitals, while Nath and Sudharshan (1994) analyze strategy coherence in acute care. Other studies have examined diversification in transplant centers (Parker-Lue and Lieberman, 2020), competitive technology adoption (Sundaresan et al., 2023), and cross-service synergies (Huesch, 2013). These precedents establish hospitals as insightful contexts for exploring strategic choices. Our contribution lies in examining how variation in the speed of information access affects decisions and outcomes in a highly competitive healthcare environment. In doing so, we also contribute to research showing that strategy is realized through the practical execution of daily micro-level decisions in organizations (Lee and Park, 2024; Jalonen et al., 2018; Healey and Hodgkinson, 2024). As Lee and Park (2024) highlight, our empirical context illustrates how organizational strategy hinges on frontline employees enacting those objectives in their decisions.

3.1.1 Data and decision structure

Between March and October of 2019 and 2022, the Emergency Department at FVL recorded over 64,000 patient encounters and more than 43,000 unique patients.

We focus on non-critical, non–time-sensitive patients—those requiring urgent physician evaluation but not in immediate danger—for whom diagnostic test results remain clinically informative over the several-hour decision window. Upon arrival, patients are screened by administrative staff and a triage nurse: critical cases (e.g., trauma, heart attacks) are sent directly to resuscitation, minor cases are dismissed, and the remaining patients proceed to physician evaluation. For these non-critical patients, Emergency Department length of stay is largely determined by when physicians resolve diagnostic uncertainty and decide whether to discharge or hospitalize, rather than by treatment protocols or resource constraints that dominate length of stay for critical cases. The remaining patients are assigned to one of two Emergency Department wards based on insurance status and triage level: the regular ward (national insurance, triage levels 1–3) or the private insurance ward (private insurance, triage levels 1–5).[6]

Each patient typically interacts with two physicians. The initial consulting physician gathers the patient's history and symptoms and orders diagnostic tests. The patient is then moved to a bed and cared for by an attending physician, who monitors progress, reviews test results, and decides whether to order more tests, continue observation, discharge the patient, or admit them to the hospital. We observe 387 attending physicians who made these final decisions.

We observe the full sequence of each decision: physician characteristics, patient condition (e.g., vitals, triage), the number and timing of lab tests ordered,[7] the time spent, and the decision outcomes. Beyond the number of laboratory tests requested, our outcome measures include the total charges billed to the insurer, length of stay (a proxy for decision speed), patient satisfaction, whether the patient was hospitalized, whether they returned to the ER within 30 days.[8]

The outcomes we track map directly onto the dimensions of strategic positioning in healthcare,

[6] Triage levels indicate clinical urgency, with level 1 being the most severe.

[7] The tests include blood counts, metabolic panels, urinalyses, and many others performed in-house.

[8] Charges include all billed items between ER admission and discharge: tests, drugs, consumables, and physician fees.

capturing more than just operational efficiency. Satisfaction and clinical outcomes are common indicators of quality differentiation, capturing both patient perceptions and health results. However, speed and responsiveness—such as length of patient stay—constitute a distinct and increasingly salient dimension. For many patients, particularly in competitive settings, how quickly care is delivered is a key factor in their choice of provider. In contrast, charges reflect internal efficiency and throughput, and can drive a more value-based positioning. These decisions shape how hospitals strategically position themselves—whether by prioritizing speed or quality, minimizing charges, or attempting to balance all three.

Interviews with physicians and administrators confirmed that these metrics are strategically salient for the hospital. They highlighted that these attributes influence how they attract patients, negotiate with insurers, allocate resources, and monitor performance. Administrators noted that in particular, high-paying or well-insured patients often switch hospitals if dissatisfied, and emergency wards are acutely aware of these dynamics. These pressures influence how hospitals allocate resources, and highlights why faster access to information can matter strategically.[9]

Increasing the speed of information access In June 2022, FVL introduced a technology that accelerated physicians' access to test results in one of the two wards, by reducing delays in detecting result availability. This change provided a rare opportunity to examine how faster access to information alters decision-making. Before the intervention, attending physicians had to repeatedly log into a separate, password-protected system to check whether test results were available. This non-delegable task required launching the application, entering credentials, manually searching for the patient—most commonly by entering a 7–11 digit national identification number—and sequentially opening each test group to verify result availability across multiple screens. Given that the average episode involves 4.68 laboratory tests, and substantially more at the upper tail, repeated access checks could conservatively consume between 10 and 24 minutes of physician time per episode,

[9] Highlighting the importance of speed in this competitive context, the Chief Laboratory Officer of the hospital remarked: "What led the hospital to undertake this pilot was the need to be more efficient in the ER, which in Colombia is an activity that generates economic losses. The most important variable is time. EDs frequently have over-occupancy of 120 to 150% or more." (Chief Laboratory Officer)

leading to frequent delays.[10]

Specifically, a new dashboard was installed in the private insurance ward. Developed in collaboration with a local software firm and the research team, the dashboard continuously displayed the status of all tests ordered for each patient in color-coded circles, without showing the numeric test values.[11] While physicians still needed to log in to view the numerical values, they could immediately see which results were ready. The system offered passive visual information only—no pop-ups, no sounds, and no push-based alerts –see Figure A.1–.

The dashboard was installed in the third week of June 2022, with physician training completed by early July. We therefore define July 2022 as the start of the treatment period.[12] This implementation created an environment where two nearly identical wards differed only in the speed at which attending physicians could access test results. This variation allows us to identify the effect of faster information access on information acquisition and the speed and quality of decisions. A notable feature of our setting is that the lab directly contacts the physician when a test result is critical, sharing the exact value, to prompt closer attention to the patient. Importantly, this process remained unchanged before and after the technology was introduced and was consistent across wards, so does not affect the interpretation of our empirical approach.

We highlight that organizations can increase the speed of information access through many channels beyond dashboards (e.g., alerts, automated detection, workflow and process redesign), and

[10]Figure A.2 shows that observable characteristics of the ward and patients—such as time of day, day of week, or patient demographics—are weak predictors of test processing time (with R^2 values never exceeding 0.085). This implies that physicians face substantial uncertainty about when test results will become available and cannot reliably predict when to check for them. On average, laboratory results take approximately two hours to be processed, with no statistically significant differences across wards or between the pre- and post-intervention periods.

[11]Each time a new shift of doctors began, they were required to log into the dashboard; similarly, they logged out at the end of their shift. Logging in was necessary for the dashboard to receive and display information. The dashboard used color-coded circles to indicate the status of lab tests: white for pending, blue for validating, green for completed and normal, yellow for concerning, and red for critical.

[12] A related concern is whether the dashboard improved decisions by increasing the *completeness* of information rather than the speed of access. We find this interpretation unlikely. The dashboard did not provide physicians with any additional or more complete information: the set of diagnostic tests they could order, laboratory processing, staffing, and clinical protocols governing discharge and hospitalization decisions were unchanged across wards and periods, and urgent results continued to be communicated directly by phone. Moreover, we know empirically that in the majority of episodes—across wards and periods—patients are discharged only after the last ordered test result has been released. This indicates that physicians already waited for all requested diagnostic information before making final decisions prior to the dashboard. Thus, the intervention did not mechanically increase information completeness at the time of decision.

we remain agnostic about the specific implementation. In this particular case, reducing frictions to access information via this dashboard increased the speed of information access, but this may not always be the case. For example, removing logins for the hospital's existing system without any visual display of when test results are available could reduce access frictions yet still fail to meaningfully accelerate when physicians view results. Similarly, increasing salience may often speed up information access, but not necessarily: for example, if many items are highlighted or alarms fire frequently, users may habituate and salience can lose its signaling value. Accordingly, we focus on the consequences of accelerating when decision-makers access information, regardless of how that acceleration is achieved.

We also distinguish how faster information access is related to attention. Information cannot be processed without being attended to, and attention is therefore a necessary component of information access. However, the dashboard does not change *what* physicians attend to or *whether* they attend to test results—those decisions were already governed by clinical protocols and professional norms. Instead, it changes *when* information becomes visible and therefore *when* it can enter the attention and decision process. In this sense, the intervention affects the *timing* of attention rather than the allocation of attention across information sources.

Importantly, both the empirical evidence and our conversations with physicians indicate that diagnostic information does not depreciate over the decision horizon we study. Physicians do not review results so late that they become clinically obsolete; instead, results are typically accessed within a timeframe spanning hours rather than days or weeks—during which patients' conditions and symptoms remain stable. Acute and highly time-sensitive cases (e.g., cardiac events or severe trauma), where minute-by-minute information decay is critical, are triaged to the resuscitation unit and are not included in our empirical analysis. For the non-critical cases we study (e.g., non-urgent COVID cases, minor injuries, sprains), interviewed physicians report that diagnostic test results remain clinically relevant and actionable throughout the patient's length of stay. Thus, while faster access to information can in principle reduce depreciation in other settings, information decay is unlikely to be a primary mechanism in this context.

4 Empirical strategy and data

In this section, we outline our empirical strategy and discuss the rationale behind the use of a triple-differences specification. We also explicitly outline our method for testing the parallel trends assumption and provide the main descriptive statistics of our data. Finally, we present arguments supporting the validity of our research design.

Differences-in-differences Our empirical strategy leverages a differences-in-differences (DiD) specification. Cases assigned to the private ward, where the technology was implemented from June 2022 onwards, represent the treatment group. Cases assigned to the regular ward, where the technology was never set up, represent the control group. Specifically, a baseline difference-in-differences model would estimate:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)}) + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \gamma' \mathbf{X_i} + \epsilon_i \tag{1}$$

where y_i is an outcome (such as length of stay) of case i, $w(i)$ indexes the ward to which patient i is assigned, $d(i)$ indexes the physician allocated to patient i, and $t(i)$ indexes the exact hour (i.e. date and hour of day combination) in which patient i arrived. The model controls for physician $\alpha_{d(i)}$, insurance company $\theta_{in(i)}$ and hour $\pi_{t(i)}$ fixed effects, as well as patients' pre-determined characteristics $\mathbf{X_i}$ (patient age and gender, triage level, main diagnosis and patient vital signs on arrival to the ED).[13] Our model includes insurance company and physician fixed effects, leveraging a key feature of the hospital: medical staff are regularly rotated across wards.[14] This allows us to isolate the effect of the technology from physician-specific factors by comparing the same doctors across different settings.[15] The parameter β captures the average differential outcome for cases

[13] Instead of controlling for the ward to which the patient is assigned, we control more finely for the detailed insurance company of the patient as the insurance company fully determines the ward.

[14] Length of stay in an emergency department depends on a variety of factors, including the number of patients present, the laboratory's workload (i.e., the number of tests being processed), the availability of doctors, and the characteristics of the patient's condition. To account for the first three factors, we include detailed hour-by-hour fixed effects. To control for patient-level variation, we include the full set of observable characteristics measured at the time of the patient's arrival. In addition, we control for the identity of the attending physician.

[15] A small number of physicians appear only in one ward, typically because they spend limited time in the hospital. These cases do not contribute to the estimation, as the physician and insurance fixed effects are perfectly collinear.

assigned to the private insurance ward following the introduction of the technology.

Triple-differences strategy In the context of Emergency Departments, the exploitation of organizational changes to one hospital ward while using a different ward as a control group in a DiD strategy was pioneered by Chan (2016).[16] The identification assumption requires that the average outcomes across the two wards would have evolved similarly in the absence of the introduction of the technology.

A challenge in our setting is that, according to our discussions with FVL administrators and physicians, the introduction of the technology potentially coincided with seasonal changes in the composition of cases. Specifically, it may be that healthier patients (even after controlling for patient characteristics) may reach the private insurance ward in the summer months, relative to the winter months and to the regular ward. To alleviate this concern, our main estimating strategy is a triple-differences model comparing the months after June in the private insurance ward in 2022, relative to the regular ward and to 2019, the last pre-COVID year.[17] Specifically, we pool the 2019 and 2022 March-October months together in the sample and use as the main independent variable of interest the triple interaction between $(Private_{w(i)} \times Post_{t(i)})$ and a year 2022 dummy. The model becomes:

$$\begin{aligned} y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i. \end{aligned} \tag{2}$$

In this triple-differences model, all the control variables (i.e., $\mathbf{X_i}$) are interacted with the year 2022 dummy to capture different evolution in the baseline characteristics of the patients.

A key concern for identification is that patient or physician characteristics might have changed differently across wards around the time the technology was introduced. Figure A.3 directly ad-

[16] We follow Chan (2016) in clustering the standard errors at the physician level.

[17] We chose the year 2019 as it is the last pre-COVID year. In 2020 and 2021 multiple changes to the internal organization of the ED (including the temporary elimination of the private insurance ward) make comparisons difficult. If instead, we choose 2018 instead of 2019, we obtain very similar estimates.

dresses this concern by reporting triple-difference estimates for a wide range of observable patient and physician characteristics. We find no evidence of differential changes in patient composition, illness severity at admission, or physician characteristics coinciding with the rollout of the dashboard. In particular, triage categories and indicators of extreme vital signs remain stable around the introduction date. These patterns indicate that the estimated effects are not driven by shifts in the types of patients treated or their initial severity, but instead reflect changes in behavior following the technology adoption.[18]

The main tables report the results when the continuous (Length of Stay and Charges) and count measures (number of laboratory tests) are log-transformed. We do so to minimize the impact that outliers can have on our results. All our results are consistent and qualitatively similar if instead of using the log-transformation, we use the Inverse Hyperbolic Sine (IHS) transformation.

Event study analysis We assess the difference-in-differences identification assumption by testing for differential pre-trends using the following leads-and-lags model:

$$y_i = \sum_{j=-K...-1}^{1...K} \beta_j (Private_{w(i)} \times Monthj_{t(i)}) + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \gamma' \mathbf{X_i} + \epsilon_i \tag{3}$$

where $\hat{\beta}_{-K}, ..., \hat{\beta}_{-1}$ capture the estimated effects of being assigned to the private insurance ward in the K months leading to the introduction of the technology, and $\hat{\beta}_1, ..., \hat{\beta}_K$ capture the corresponding effects for the K months following the introduction in June 2022.[19]

Descriptive statistics Our main analysis sample comprises eight months centered around the introduction of the technology in June 2022, and their equivalent in 2019. The $64,152$ cases in our

[18]Two observable characteristics—patient gender (male) and age—exhibit small negative and statistically significant coefficients. This pattern indicates a modest increase in the share of younger male patients in the private ward following the introduction of the technology. To address any potential concerns related to changes in patient composition, all subsequent analyses explicitly control for patient age and gender. As a result, the estimates we report are robust to these compositional shifts.

[19]Note that our empirical specification is not affected by recent criticisms about DiD designs (de Chaisemartin and D'Haultfeuille, 2017; Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021). First, treatment is not 'fuzzy' as defined in de Chaisemartin and D'Haultfeuille (2017) because no case is treated in the control group. Second, the treatment affects all the (treated) cases simultaneously and the private insurance ward remains treated for the remainder of the sample period. This rules out the concerns related to staggered treatment designs (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021).

sample include 43,607 distinct patients cared for by 387 distinct physicians.

Table A.2 presents summary statistics for the main variables, organized into four panels: by period (before and after the dashboard introduction) and by ward type (private vs. non private). We examine three sets of outcomes. First, the number of laboratory tests ordered and associated total charges test *Hypothesis 1*, concerning whether faster information access leads to less and more targeted information acquisition. Second, patients' length of stay tests *Hypothesis 2*, concerning whether decisions are reached more quickly. Third, hospitalization, readmission, and patient satisfaction test *Hypothesis 3*, concerning whether decision quality improves. For more details of the main variables used in this paper, refer to Appendix Section A.2.

In the private ward, prior to the dashboard, the outcome variables exhibit a right-skewed distribution. For example, the average length of stay is 0.4 days, which is notably higher than the median of 0.17 days. This pattern of right-skewness is consistent across all four panels. In the same panel, the average patient undergoes 4.6 laboratory tests, the average logged episode charge is 13.1, 15% of patients are hospitalized, and 12% return to the emergency department within 30 days.

Comparing wards in the pre-dashboard period, patients in the private ward have fewer tests, lower charges, lower average length of stay, and lower hospitalization rates than those in the non private ward. Readmission rates are similar across wards. These patterns suggest that, even before the intervention, the private ward likely served healthier patients.[20]

Looking at variation over time, we see that the private ward shows little change in outcomes after the dashboard was introduced. In contrast, the non-private ward experiences noticeable increases in length of stay, number of tests, charges, and hospitalizations. According to hospital staff, these variables typically rise around July—the month immediately following the introduction of the technology—reflecting recurring annual cycles observed over multiple years. This seasonal

[20]Table A.3 reports summary statistics for patient gender, age, and triage level. These characteristics remain relatively stable over time within each ward. Notably, in the pre-dashboard period, the private ward admits fewer triage 1 and 2 patients—the most severe cases—suggesting that patients in this ward are, on average, healthier. We control for these observables in all of our estimations.

pattern is precisely why our main econometric specification relies on a triple-difference design: we compare the post-June months in the private ward in 2022 not only to the non-private ward in the same year, but also to the same period in 2019, before the dashboard existed.

The absence of such increases in the private ward suggests that the introduction of the dashboard may have had a mitigating effect, and that this effect is visible even in the raw data. These raw patterns do not undermine our interpretation of the main results. Rather, the observed decreases in the private ward should be understood relative to the counterfactual of the non-private ward: without the dashboard, the private ward would almost certainly have shown similar increases.[21]

5 Information access speed and information acquisition

In this section, we display results from testing *Hypothesis 1* concerning the impact of information access speed on information acquisition, by examining the tests that physicians ordered—how they acquire information in this context—and the total associated charges.

5.1 Main results on information acquisition

In Column 4 of Table 1, we examine whether physicians changed the number of laboratory tests that they ordered (according to equation 2). In this table and the ones that follow, standard errors are reported in parentheses. We find a precisely estimated negative coefficient: on average, the number of tests ordered decreased by 10% following the introduction of the technology. This finding suggests that increasing the speed of information led to less information acquired, decreasing the amount of overtesting as defined by the Chief Laboratory Officer at the hospital: "when a result is not relevant, it is lab test overutilization."

—— *Insert Table 1 Here* ——

We confirm these findings in Figure 1, where, we find that the number of tests ordered remained broadly unchanged in the months leading up to the introduction of the technology, and then

[21] Table A.1 shows that increases in length of stay in the control ward over this calendar window are common and not unique to 2022, reflecting a recurring seasonal pattern observed in several pre-treatment years. By contrast, after the dashboard's introduction, this seasonal increase does not occur in the treated ward, leading to a post-period divergence driven by the treated ward deviating from its historical pattern.

decreased around 20% in August 2022.

—— *Insert Figure 1 Here* ——

Furthermore, Rambachan and Roth (2023) provides insight on how to conduct robust inference in designs that use difference-in-differences designs even if the assumption of parallel trends' assumption may be violated. The idea is that we can estimate how large the post-treatment violations of the parallel trends have to be to make the design invalid. Figure A.4 conducts this robust inference –for this and the outcomes below– and shows that the violation of the parallel trends assumption has to be very large (larger than it could be, given Figure 1 and Table A.2) for our suggested effects to be invalid. Our results (here and for the other dependent variables) indicate that the estimated effect is robust to small and moderate violations of the parallel trends assumption. The event study figures show no significant differences before the technology was introduced, supporting this assumption. Simulations further suggest that even with moderate deviations from parallel trends (e.g., 30% or 80%), the results remain statistically significant. However, when deviations become large (e.g., 130% or 180%), the estimates lose significance and are no longer robust. Importantly, our event study does not indicate any such large pre-treatment differences. The evidence from Figure 1 and A.4 provide further validity to our design, suggesting that the difference between the two emergency department wards remained similar before the intervention but not after.

The econometric result–that faster information access led to less information acquisition–is echoed in our qualitative interviews. Physicians felt that having faster access to the right information allowed them to rely on fewer tests overall. For example, one physician said, "*This screen might be good for the hospital, because getting information faster makes me think twice about whether I really need to order the next test.*"

Furthermore, we evaluate *how* information acquisition processes changed by exploring heterogeneity along two dimensions. First, we take advantage of the distinction between general tests—used across a wide range of conditions—and specialized tests, which are more targeted. Second, we consider the timing of test orders within a patient episode, distinguishing between those typically ordered early in the process and those more commonly ordered later.

Specialized and generic tests Generic tests are those that are frequently used across a wide range of conditions. For instance, Complete Blood Count (CBC) measures the levels of white and red blood cells as well as platelets. These tests are very generic as they are commonly ordered to assess overall health and detect conditions such as anaemia, infection, and leukaemia. Another example is the Urinalysis, which analyzes urine for various substances that can indicate different conditions, such as such as urinary tract infections, kidney disease, and diabetes. Other tests are more specific and targeted, such as the Angiotensin Converting Enzyme (ACE) test, which is used to monitor patients with diagnosed sarcoidosis—a rare disease.

We classify a test as generic if its category appears in 50,000 or more episodes—corresponding to the 99th percentile of the distribution—and as targeted otherwise, using all episodes from 2016–2022.[22] We find that physicians significantly reduced orders of generic tests, focusing more on targeted tests (Columns 1 and 2 in Table 2)—consistent with our first hypothesis that information access speed led to less, more targeted information acquisition.

—— *Insert Table 2 Here* ——

Timing of testing We also find that the largest reduction in the number of tests stemmed from the end of the decision, suggesting that earlier tests may have been effective in finding a promising path to diagnosis—also consistent with the hypothesis that information access speed enabled less, more targeted information acquisition for decisions. Within the sample of cases where tests were ordered, we cut the length of stay into quartiles for each case. Columns 3-6 in Table 2 show that the reductions in the number of tests ordered stem mostly from the last quartile. This reinforces the idea that physicians were able to make more relevant information acquisition choices earlier on, decreasing the need to order additional tests at the end of their decision period.

[22] Our results are robust to alternative ways of splitting the data. In particular, when we use different cutoff points—such as 3% and 5%—to define the groups, the estimated effects remain stable in magnitude and significance. Table A.4 reports these alternative splits and shows that the findings are not sensitive to the specific choice of cutoff.

5.2 Effect on Total Charges

We then explore whether these patterns of more efficient information acquisition led to lower associated charges billed to the health systems. Column 8 of Table 1 examines an outcome variable that indicates the total charges billed to the insurer (according to equation 2). We find that this total charges variable decreased by around 25% following the introduction of the technology. Given that the average charge is about 1.4K US dollars, this represents a decrease of about 365 US dollars, approximately 1.5 times the local monthly minimum wage in 2022. These results are robust to winsorizing the charge variable. For instance, when we winsorize at 95% or 90%, the coefficient is still negative and significant at 1%, estimated at 24% and 22%, respectively. It is interesting to note that the effect of the decrease in the number of tests is far below the remarkable 25% decrease in charges from Table 1, suggesting that physicians eliminated relatively more expensive tests.

Figure 1 provides evidence consistent with parallel trends in total charges. Specifically, it shows that, prior to the intervention, trends in patient charges were similar across the two wards. This suggests that any post-intervention differences in charges can more plausibly be attributed to the intervention itself, rather than to pre-existing differences in trends.

The observed reduction in total charges is also consistent with qualitative evidence from physician interviews. Several physicians noted that more targeted test ordering should translate into lower out-of-pocket expenses for patients, and therefore into higher patient satisfaction (to be confirmed below), because patients receive the necessary care without incurring the costs of unnecessary diagnostics.

Overall, we find strong support for *Hypothesis 1* that higher information access speed leads to less and more targeted information acquisition for decisions. Faster access to information reduced the number of tests ordered, especially by decreasing generic rather than specialized tests *later* in the medical episode. As a result, we also observe a corresponding reduction in total charges.

6 Information access speed and decision speed

In this section, we display the results of testing *Hypothesis 2* that increasing the speed of information access leads to a faster decision. To do so, we measure the length of time to reach a decision using the patient's length of stay in the Emergency ward, an important measure of performance for Emergency departments in hospitals (Chan, 2016, 2018; Chan and Chen, 2022). [23]

Baseline results We find that increasing the speed of information access leads to a faster decision. Table 3 displays the baseline results from running our triple-differences strategy according to equation 2. We start in Column 1 with a simplified model, which only controls for a post-June dummy and a private insurance ward dummy (both interacted with the year 2022 dummy). In Columns 2 and 3 we then sequentially introduce insurance status and hour-fixed effects, as well as (initial consultation) physician-fixed effects. From column 4 onwards, we add patient controls. Column 4 additionally winsorizes the top and bottom 1% of the length of stay distribution. While log-transforming this dependent variable should have alleviated the strong skewness of the length of stay distribution, winsorizing the top part of the distribution contributes to reassuring us that the baseline estimates are not disproportionately due to extreme positive values. We find that the introduction of patient controls and winsorizing the distribution has a strong effect on the estimate, decreasing it in absolute value from $-.35$ to $-.16$. In column 5, we drop Triage 4 and 5 cases from the sample without winsorizing the distribution. Because these cases are only present in the private insurance ward, dropping them from the sample increases the homogeneity of the average case across the two wards, but substantially decreases our sample size. Finally, column 6 adds back the Triage 4 and 5 cases without winsorizing the distribution, and the treatment estimate remains broadly similar to column 5, and is more conservative than winsorization in column 4. Given this, we take this model as our baseline estimate.[24] The reduction of 13% represents a decrease of about

[23] While treatment times may vary across conditions, we do not expect this to affect our estimates provided that both wards observe similar variation in conditions. While the two wards differ in triage levels, we control for them across our regression specifications.

[24] We obtain similar results if we winsorize the distribution of the length of stay in this column.

75 minutes with respect to the mean –of the private ward in the pre-period– (the average episode lasts about 9.6 hours).

—— *Insert Table 3 Here* ——

While emergency department length of stay reflects multiple operational factors, it is a particularly informative measure of decision timing for non-critical patients, for whom exit from the ED is primarily determined by the physician's discharge or hospital admission decision. In our setting, staffing levels, bed availability, and diagnostic processing times did not change with the introduction of the dashboard, and we find no evidence of shifts in patient severity or composition at admission (Figure A.3). We therefore interpret reductions in length of stay below as reflecting earlier resolution of diagnostic uncertainty and faster disposition decisions, rather than changes in downstream capacity constraints.[25]

To better isolate the impact of the dashboard introduction, we further examine whether the reduction in length of stay is driven by urgent or non-urgent cases.[26] We expect the dashboard to have a stronger effect on non-urgent cases, since in urgent cases the laboratory directly contacts the physician, and therefore should not be impacted by the reduced information access time introduced by the dashboard. Table A.6 is aligned to this hypothesis –although with noisy estimates for both coefficients–: the effect is negative for non-urgent cases and non-negative for urgent cases..[27]

We display in Panel C of Figure 1 the estimates of equation (3) for Length of Stay as dependent variable. We do not find a significant differential effect of being assigned to the private insurance ward in the months leading to the introduction of the technology. In the first month period after this introduction, a discontinuous decrease in length of stay of around 35% is apparent in the

[25] To better understand the magnitude of these estimates and provide further validation to our triple-differences model, Table A.5 displays the baseline results from estimating Equation (1). We find consistent results. The DiD model centered around July 2022 suggesting in Column 1 that the introduction of the technology is associated with a 25% decrease in the average length of stay. In Column 2, we repeat the estimation of (1) but in the 'placebo' year of 2019, in which no technology was introduced at any point. We find that the post-June period was associated with a decrease in length of stay of around 11%. While the decrease in the post-June months is much smaller in 2019 relative to 2022, it is still statistically significant. This suggests that there might be seasonal effects in the differential composition of patients arriving to the private insurance and regular wards. This confirms that a triple-differences model is more likely to be appropriate in our setting.

[26] An episode is classified as urgent if at least one test result falls outside the defined criticality range. The number of such cases is low, as most test results lie within standard clinical thresholds.

[27] See below Table A.13 for complementary evidence in support of this interpretation.

figure. Overall, we interpret the evidence in Figure 1 as largely supportive of the main identification strategy in the paper.[28]

Decomposing the effect of information access speed on decision speed This subsection decomposes the effect of information access speed on length of stay.

First, the dashboard eliminated repeated access costs by passively displaying test readiness in real time for all patients in the ward. Our triple-differences estimates indicate an approximately 13% reduction in length of stay, which corresponds to about 75 minutes relative to a baseline mean of 0.4 days (576 minutes). Back-of-the-envelope calculations suggest that the mechanical time savings from avoiding repeated manual checks—roughly 10–24 minutes per episode—can account for only about one-sixth to one-third of this total effect. This implies that most of the reduction in length of stay operates through changes in how physicians sequence information acquisition and resolve uncertainty, rather than through mechanical effects.

We also explore how much of the decrease in length of stay might be driven by the reduction in tests acquired, by comparing cases with and without laboratory testing. The rationale is that if faster information access affects decisions by changing information-gathering behavior, its effects should be concentrated in cases that require tests and minimal in cases that do not.

Cases with and without tests We investigate how much of the increased speed in decisions arose from more efficient information acquisition, by exploring whether these effects were driven by cases that required ordering tests. If information access speed increases the speed of decisions through the mechanism of changing information acquisition choices, then we should find that cases that did not involve acquiring further information through laboratory tests (e.g., a sprained ankle) should not be affected strongly by the introduction of the technology. Indeed, we find consistent

[28] As shown in Table A.2, the average length of stay in the raw data is lower in the private ward. However, Figure 1 suggests a possible positive difference in the pre-period, indicating a higher average length of stay in the private ward. This apparent discrepancy arises because the figure incorporates a rich set of controls, allowing for a more precise and comparable analysis across individuals.

evidence (see Table A.7 and Table A.8).[29]

As an alternative approach, we use the main diagnosis of each episode to assess how frequently laboratory tests were ordered for that condition in the *past*. This approach relies on variation that is less directly influenced by the physician's personal decision-making, in contrast to analyses based solely on whether tests were ordered—since that decision is made by the physician. Table A.9 presents the results of this analysis, which are fully consistent with those reported in the previous tables. We find that episodes involving diagnoses historically associated with frequent test ordering show the largest reductions in test use following the increase in information speed. The technology has a strong negative effect on test use for these diagnoses. In contrast, for diagnoses where lab tests are rarely ordered, the effect is close to zero.

To better understand how much this efficiency in information acquisition mediates the relationship between changes in information speed and length of stay, we conduct a mediation analysis following Heckman and Pinto (2015), the details of which we report in Section A.1. Our findings show that the mediator ratio is 48%, indicating that about half of the treatment effect stems from physicians' ordering fewer tests.

Overall, the treatment reduces the length of stay by 75 minutes. Of this effect, 48% is attributable to fewer tests being ordered. The mechanical reduction in length of stay associated with fewer tests is approximately 17 minutes—the average of 10 and 24 minutes—which accounts for about 22% of the total effect. The remaining share of the mediated effect implies that roughly 30% of the total reduction (about 22 minutes) is driven by the ability to learn better from the information acquired.

In summary, these findings provide robust evidence in support of *Hypothesis 2*, showing that increasing the speed of information access leads to faster decisions. This effect may alone in some

[29] Table A.7 shows that the length of stay indeed decreases for cases in which tests were ordered, and not for cases in which they were not. Column 2 shows a slightly negative but statistically insignificant effect of the technology cases that do not involve ordering tests. However, the negative effect is precisely estimated and larger in magnitude when we focus on cases for which physicians order tests (column 3). In Table A.8, we find that the length of stay decreases by about 20%, the number of laboratory tests by 20%, the total charges by 17% and the probability of hospitalization by 4% for patients with a diagnosis that require tests *vis a vis* other patients with a diagnosis that do not require laboratory tests.

cases provide advantages for organizations, enabling them to make decisions before their competition, and suggests that one key lever through which organizations can realize speed advantages is through investing in the speed of information access.

7 Information access speed and decision quality

In this section, we empirically investigate *Hypothesis 3* on how the speed of information access affects decision quality, focusing on the outcomes related to quality of care and patient satisfaction. This helps us directly assess whether less information acquired and faster decisions could reflect premature closure, and investigate whether information access speeds leads to improved decisions.

Hospitalization and Reincidence In Column 3 of Panel A of Table 4, we find a large effect on the quality of care, decreasing the likelihood that the patient is hospitalized, rather than being discharged home. The decrease in this likelihood is around 7 percentage points, which represents around 27% of the unconditional likelihood. To the extent that admission to the hospital represents an acknowledgment that the patient has not improved sufficiently during her stay at the Emergency ward, we can conclude that increasing the speed of information access through the introduction of the technology improved patient outcomes.[30] In the context of overcrowded emergency departments, a decrease in hospitalization also typically implies less patient boarding.[31] Second, we find in Columns 4-6 of Table 4 Panel A that this decrease in hospitalization happened without increasing the likelihood that the patient would return to the Emergency Department in the next month. The estimate is close to zero and is statistically insignificant, suggesting a safe discharge and thus an overall improvement in the quality of care.

[30] Conditional on two patients with the same initial condition (i.e., same initial vital signs, triage, entry time, physician), hospitalization is likely to reflect a worse treatment for the condition than being discharged home since it means that the patient was not able to recover sufficiently.

[31] Boarding is keeping patients admitted to the Emergency Department for long periods due to inadequate capacity of inpatient wards. Boarding has been associated with negative effects on mortality and morbidity (Sartini et al., 2022). The pooled estimate in Table 4 compares average outcomes before and after the intervention. Panel D of Figure 1 shows that post-intervention coefficients are consistently negative, while pre-intervention coefficients are near zero but positive. Although individual post-period estimates in the Figure are imprecise, their systematic contrast with the pre-period yields a statistically significant pooled effect in the Table, indicating a reduction in hospitalizations distributed across the post-intervention months.

—— *Insert Table 4 Here* ——

The leads and lags figures for these outcome variables can be found in Figure 1. For all outcomes, the graph shows no significant differences between the private insurance ward and the other ward in the months leading up to the introduction of the technology, providing further support for our findings. Moreover, Figure A.4 shows that the potential for violation of the parallel trend assumptions that would invalidate the research approach is much larger than they are according to Figure 1, providing additional support to the validity of our econometric design.

Patient satisfaction We additionally find that the introduction of the dashboard improved patient satisfaction with the care provided in the private insurance ward, relative to the regular ward (Table 4 Panel B).

Our measures of patient satisfaction are based on a survey sent to a randomly-selected subset of patients. We focus on three responses to the survey: (1) whether the patient thought that the medical staff displayed the right attitude in the provision of care, (2) whether the patient thought that the physician complied with good medical practices, and (3) whether the patient believed that the physician was good at answering questions and addressing potential concerns. Unfortunately, we only have these survey results for the year 2022 and for a sample size of fewer than two thousand observations. As a result, the specification for these variables is a DiD model that follows equation 1 (with coarser time effects), and results are noisier.

Panel B of Table 4 displays the results of these estimations. We find that the introduction of the technology had positive effects on patient satisfaction for all three measures. In terms of magnitude, the improvement on the attitude question is 0.2, which represents 37% of the 0.54 sample standard deviation. The increase in compliance and reported willingness to answer questions are slightly smaller at around 26% and 31%, respectively. We also use as a dependent variable the average of the three questions and find qualitatively similar results. The effects are large in magnitude, which illustrates how patients perceived the improvement in decisions resulting from increasing information access speed.

In Figure A.5, we plot the estimated leads and lags from equation (1) of patient satisfaction. We find that patient satisfaction is not trending in any direction before July 2022, after which there is a meaningful improvement in all three measures as well as the average. Despite some noise in the estimates, Figures A.5 and A.4 (Honest DiD for the average of the three questions) suggest that the dashboard speeding up information access impacted patient satisfaction.

Information Access Speed and the Cognitive Load Mechanism To explore what might drive these improvements in quality of care and patient satisfaction, we explore the cognitive channel raised in Section 2, that the speed of information access may enhance decision quality by enabling decision-makers to process the information more effectively. These analyses are meant to be exploratory in nature, rather than providing indicative evidence for this mechanism.

To do so, we investigate two dimensions of heterogeneity: whether the ward was busy, and whether the diagnosis was common or uncommon. If information access speed affects decision quality by enabling decision-makers to better cognitively process the information, we would expect that faster information access should have a stronger effect when cognitive demands are higher. This could be the case during busy times, when physicians experience higher cognitive load due to having to switch between patients, which may reduce their ability to process and learn from information. It could also be the case for less common diagnoses, where lack of experience or familiarity requires physicians to think harder and spend more time reasoning through potential hypotheses and actions.

Ward busyness We find that the introduction of the technology indeed had a larger effect during busy hours (Table A.10).[32] For each hour-ward combination, we construct the ratio of patients per physician and label the ward as busy if the ratio is larger than 2, which also represents the 75th percentile of the distribution of patients per physician. Our interpretation of a busy ward is that physicians had more pressure to switch from one patient to another patient. When physicians

[32]Unfortunately, data limitations prevent us from examining patient satisfaction outcomes in this and other heterogeneity analyses.

need to switch across patients, it is likely more difficult to cognitively hold all details of each patient—whereas during slow times, physicians see a single patient and it can be easy to remember and hold all details of that patient, regardless of the amount of time that has passed for processing the result. For all of our outcome variables, the magnitude of the treatment estimate during busy hours is nearly always more than double the estimate during non-busy hours.[33] This provides suggestive, explorative evidence that faster information access may have enabled them to make better decisions by allowing them to better cognitively process the information when operating under constraints.[34]

Common and uncommon diagnoses We also examine whether the impact of the technology differed depending on how common the patient's case was—which determines the amount of cognitive effort needed to make inferences and process the information acquired. If increasing information access speed helps decision-makers better process the information, we would expect that the effects should stem mainly from uncommon cases, which require physicians to think carefully through potential hypotheses and diagnoses, compared to more common cases where the diagnoses and necessary tests should be more routine and straightforward.

To investigate, we take the distribution of the universe of diagnoses of patients admitted to the hospital during the period 2016-2022. We label a diagnosis as uncommon if it belongs to the lowest 20th percentile of the distribution, and also examine robustness when cutting at the median. This distribution is heavily concentrated as a few diagnoses represent a large percentage of the population. Table A.10 shows that the effects of introducing the dashboard on the length of stay mainly occur in uncommon diagnoses when physicians must think through the information more carefully to make their decision, providing suggestive exploratory evidence.

Taken together, the results in this subsection show that increasing the speed of information access is associated with improved decision outcomes: higher quality of care and higher patient

[33] With the only exception of the total number of tests, which is only modestly larger in magnitude.

[34] It is also possible that the technology has a larger effect during busy times because it is precisely in these times that physicians may especially have long lags in checking test results, which can reduce the length of stay. However, this does not fully explain why we would observe larger improvements in other outcomes like quality of care.

satisfaction. We also find suggestive evidence that this may be driven by a cognitive channel, where faster information access enables decision-makers to process and learn from the information more effectively.

8 Alternative Explanations

In this section, we consider alternative possibilities to help interpret our results. First, physicians might be changing their behavior due to peer pressure that may become more salient from the introduction of the dashboard. Second, the quicker decisions—reflected in shorter patient stays—might be due to the lab processing test results faster for the private ward. Third, we examine whether the impact of the change in information access speed on outcomes may be driven by heterogeneity in physician characteristics. Finally, we explore whether the dashboard changed physicians' prioritization of urgent cases, mechanically leading to faster and improved decisions.

Peer effects A potential alternative explanation is that the dashboard, by making test result statuses visible, could motivate physicians through peer effects—by increasing performance pressure when others are present. To explore, we compare treatment effects in shifts with multiple physicians present versus those with a single physician on duty. If peer effects were a primary driver, we would expect stronger effects when more than one physician is working. However, as shown in Table A.11, treatment effects are stronger when only one physician is present. This suggests that peer effects in the form of performance pressure are unlikely to be the main mechanism.[35]

Reaction of the laboratory test unit We investigate whether our findings can be attributed to expedited laboratory test processing for patients in the treated private ward. Our partner organization maintains that this was not a contributing factor. To verify, we conduct triple-difference

[35] Instead, based on interviews with physicians, we interpret this pattern as reflecting the collaborative nature of this setting: when two physicians are present, they often share responsibilities and rely on each other to monitor lab results. For example, when a physician receives a call from the lab about a critical result, they may also inquire about results for other physicians' patients. As a result, the presence of a second physician can diffuse individual responsibility for checking the dashboard. In contrast, when working alone, physicians must monitor all results themselves, leading to greater engagement with the dashboard and larger observed effects.

estimations using the duration of test processing as the dependent variable. The results, presented in Table A.12, indicate that the laboratory did not accelerate the processing of tests ordered for the private ward following the implementation of the dashboard *vis a vis* the processing of tests ordered for the non-private ward. This suggests that the observed effect on our measure of decision speed, length of stay, is unlikely to be driven by changes in laboratory test processing times.

Heterogeneity by Physician Characteristics We further explore whether physician heterogeneity might drive our results by exploring treatment effects by their gender and age. Figure A.6 shows the estimated effects for our key outcomes, broken down by gender and experience. We do not find any statistically significant evidence that male doctors (compared to female doctors) or younger (vs. older) doctors were more affected by the change in how quickly information was delivered.[36]

Urgency and prioritization Because the dashboard also displayed whether test results were clinically urgent, one concern is that it may have altered physicians' prioritization of urgent cases. To examine this possibility, we compare changes in Length of Stay for episodes involving urgent laboratory results before and after the introduction of the screen. To complement Table A.6 and as shown in Table A.13, episodes with urgent test results do not experience a differential reduction in Length of Stay following the introduction of the dashboard. This finding indicates that, although severity information was displayed, the technology did not meaningfully affect physicians' prioritization behavior. This is consistent with institutional features of the setting: urgent results were already directly communicated by the laboratory in both periods, and the dashboard did not change clinical alerting protocols.

[36]Doctors were classified as "Young" or "Old" based on their average age between 2019 and 2022. Those with an average age above the median were labeled "Old," and those below the median, "Young."

9 Discussion and conclusion

In this paper, we explore how faster information access impacts how decision-makers acquire information and make decisions. We find that faster access results in more efficient information acquisition choices, with physicians ordering fewer and more targeted tests earlier in the decision process and reducing additional testing later on. Furthermore, we find that faster access enables not only quicker but also improved decisions, decreasing total charges, and ultimately improving the quality of care in the ED of a leading hospital.

Of course, our study has several limitations. For instance, we are unable to observe all aspects of physician behavior—for example, the amount of time spent with each patient or the exact time at which physicians access test results, which limits our ability to more precisely test some of our hypotheses. We view these gaps as promising areas for future research that could help refine and extend our findings.

More broadly, these findings suggest that we need to think more deeply about how organizations acquire and learn from information in the digital age. One implication of our findings is that organizations may not need to accelerate information collection itself, but rather invest in tools that reduce delays in accessing and recognizing the information they already have. Furthermore, our findings highlight the strategic importance of investing in systems that give frontline decision-makers faster access to critical information, especially in healthcare settings. Even small reductions in the time between when a diagnostic test is processed and when a physician sees the results can lead to better outcomes—not only by shortening patients' length of stay but also by helping physicians make better decisions. For hospital administrators and operations managers, this means that speeding up access to information should not be seen only as an IT upgrade or process improvement. Instead, it should be viewed as a key strategic tool to improve both efficiency and quality of care.

At a broader level,, in similarly fast-paced environments where decisions must be made quickly under uncertainty, investing in faster information access may be a powerful and scalable way for organizations to learn and adapt. If a marginal improvement in information access speed can result in substantial speed-based advantages and enable key decision-makers to improve their choices, this suggests that how organizations build in mechanisms to better process information and learn from data may increasingly become a key source of competitive advantage.

Generalizability and boundary conditions While our partner organization is broadly representative of emergency departments, our analysis focuses on a single organization, which may limit the generalizability of our results. We believe that the mechanisms we study can generalize to decisions made a cross a broader set of organizational contexts characterized by uncertainty, sequential information acquisition, and frictions in accessing available information. In such settings, faster information access—defined as reducing the time between when information becomes available and when it becomes observed and used by decision-makers—can reshape how decisions are made even when informational content itself does not change over the decision time frame.

These conditions arise in many business contexts. In product development and experimentation, faster access to early user or performance data can allow firms to refine subsequent tests and terminate unpromising directions sooner. In operations and supply chains, real-time dashboards can enable managers to identify bottlenecks or quality issues earlier, reducing unnecessary monitoring and accelerating corrective action. In multi-unit firms, performance dashboards can allow executives to detect deviations and resolve strategic uncertainty without waiting for periodic reports.

At the same time, our results are unlikely to generalize to all settings. When decisions are governed by fixed review cycles or formal rules, faster access may not change decision timing. When information arrives infrequently or in large batches, access speed may be less consequential. Finally, in environments where salient but weak signals crowd out more informative ones, or where information depreciates extremely rapidly, faster access may not improve—and may even worsen—decision-making. These boundary conditions highlight that information access speed is a consequential organizational design lever whose value depends on how information enters, and is processed within, the decision-making process.

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Figures and Tables

TABLE 1: THE IMPACT OF FASTER INFORMATION ACCESS ON INFORMATION ACQUISITION: NUMBER OF TESTS AND TOTAL CHARGES

Dependent Variable =	**Log. Number of Tests**				**Total Charges**			
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Post June X Private Ward X Year 2022	-0.071	-0.080	-0.132	-0.102	-0.256	-0.330	-0.238	-0.255
	(0.034)	(0.034)	(0.035)	(0.037)	(0.050)	(0.054)	(0.053)	(0.059)
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	No	No	Yes	Yes	No	No	Yes	Yes
Date X Hour Fixed Effects	No	No	No	Yes	No	No	No	Yes
Observations	64,051	63,916	63,916	62,650	63,948	63,813	63,813	62,546

This table reports estimates from regressions of a case's medical outcomes in the ED on the period during which the technology was introduced (i.e., after June), interacted with the ward in which it was introduced (i.e., private ward) and with a dummy for year 2022. The estimating equation is:

$$\begin{aligned} y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i. \end{aligned}$$

Columns 1–4 report results where the dependent variable is the log number of tests ordered, while Columns 5–8 report results where the dependent variable is total charges, defined as the log of the amount in Colombian Pesos (COP) charged by the hospital to the payer for each episode. Columns 1 and 5 include patient-level controls only. Columns 2 and 6 additionally include doctor fixed effects. Columns 3 and 7 further include insurance fixed effects (which subsume the assigned ward). Columns 4 and 8 include date × hour fixed effects. Patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) and fixed effects are interacted with the year 2022 dummy. Standard errors are clustered at the doctor level.

TABLE 2: THE IMPACT OF FASTER INFORMATION ACCESS ON THE INFORMATION ACQUISITION PROCESS: TARGETING AND TIMING OF TESTS

	Type of Tests		**Percentiles**			
	Targeted	**Generic**	**0–25**	**25–50**	**50–75**	**75–100**
Post June X Private Ward X Year 2022	-0.042	-0.102	-0.021	-0.117	-0.001	-2.335
	(0.032)	(0.034)	(0.034)	(0.123)	(0.326)	(0.530)
Post June X Private Ward	Yes	Yes	Yes	Yes	Yes	Yes
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	62,650	62,650	30,163	5,886	1,483	1,315

Note: This Table displays estimates of regressions of the log. of the number of tests on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. To determine whether the test is generic, we defined it as 1 (generic) if the test category appears in 50.000 (the median value) or more episodes and 0 otherwise. 50.000 episodes represent the 99% percentile of the distribution of laboratory tests. The last four columns show the effect of the interaction term on the log number of tests cutting the sample into 4 quartiles of the length of stay. For instance, the column 0-25 shows the effect of the change of the speed on information on the number of tests for the first quarter of the episode. The estimating equation is:

$$
\begin{aligned}
y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i.
\end{aligned}
$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. Standard errors are clustered at the physician level.

TABLE 3: THE IMPACT OF FASTER INFORMATION ACCESS ON DECISION SPEED: LENGTH OF STAY

Dependent Variable = Log Length of Stay	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Post June X Private Ward X Year 2022	-0.254	-0.383	-0.352	-0.163	-0.124	-0.133
	(0.063)	(0.089)	(0.085)	(0.049)	(0.059)	(0.051)
Post June X Private Ward	Yes	Yes	Yes	Yes	Yes	Yes
Patient Controls X Year 2022	No	No	No	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	No	No	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	No	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	No	Yes	Yes	Yes	Yes	Yes
Post June X Year 2022	Yes	No	No	No	No	No
Private Ward X Year 2022	Yes	No	No	No	No	No
Observations	63,025	61,750	61,612	62,618	40,830	61,600

Note: This Table displays estimates of regressions of a case's length of stay in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The unit of observation is a case i arriving at the ED. The estimating equation in Column 4 is:

$$\begin{aligned} y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i. \end{aligned}$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). All the patient controls are interacted with the year 2022 dummy. In Column 1 we display the most streamlined triple-differences model, which only includes a post dummy and a private dummy as controls, interacted with each other and with the year 2022 dummy, this regression includes all triages in its estimation and length of stay is not winsorized. In Column 2 we include the insurance status and the hour fixed effects (which subsume the year 2022 and private ward dummies), this regression includes all triages in its estimation, and the length of stay is not winsorized. In Column 3 we add the physician fixed effects, this regression includes all triages in its estimation, and length of stay is not winsorized. In Column 4 we add the patient controls and the top and bottom 1% of the length of stay distribution is winsorized, this regression includes all triage in its estimation. In Column 5 we add the patient controls and the sample includes only triage levels 1–3 and the length of stay is not winsorized. Column 6 is the baseline model for the full sample, this regression includes all triages in its estimation, and length of stay is not winsorized. Standard errors are clustered at the physician level.

TABLE 4: THE IMPACT OF FASTER INFORMATION ACCESS ON DECISION QUALITY: HOSPITALIZATION, READMISSIONS, AND SATISFACTION

PANEL A: EFFECTS ON QUALITY OF CARE

Dependent Variable =		**Hospitalization**			**30–Day Return**	
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Post June X Private Ward X Year 2022	-0.057	-0.058	-0.071	0.011	0.003	0.010
	(0.014)	(0.014)	(0.014)	(0.010)	(0.010)	(0.013)
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	No	Yes	Yes	No	Yes	Yes
Date X Hour Fixed Effects	No	No	Yes	No	No	Yes
Observations	63,916	63,916	62,650	63,916	63,916	62,650

PANEL B: EFFECTS ON PATIENT SATISFACTION

	(1) Attitude	**(2) Compliance**	**(3) Answering Questions**	**(4) Average**
Post June X Private Ward	0.201	0.156	0.180	0.171
	(0.059)	(0.064)	(0.058)	(0.057)
Patient Controls	Yes	Yes	Yes	Yes
Doctor Fixed Effects	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects	Yes	Yes	Yes	Yes
Month Fixed Effects	Yes	Yes	Yes	Yes
Mean Dep. Var.	3.681	3.603	3.650	3.625
SD Dep. Var.	0.546	0.593	0.577	0.545
Observations	1,445	1,233	1,443	1,232

Note: Panel A displays estimates of regressions of a case's different medical outcomes in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The estimating equation is:

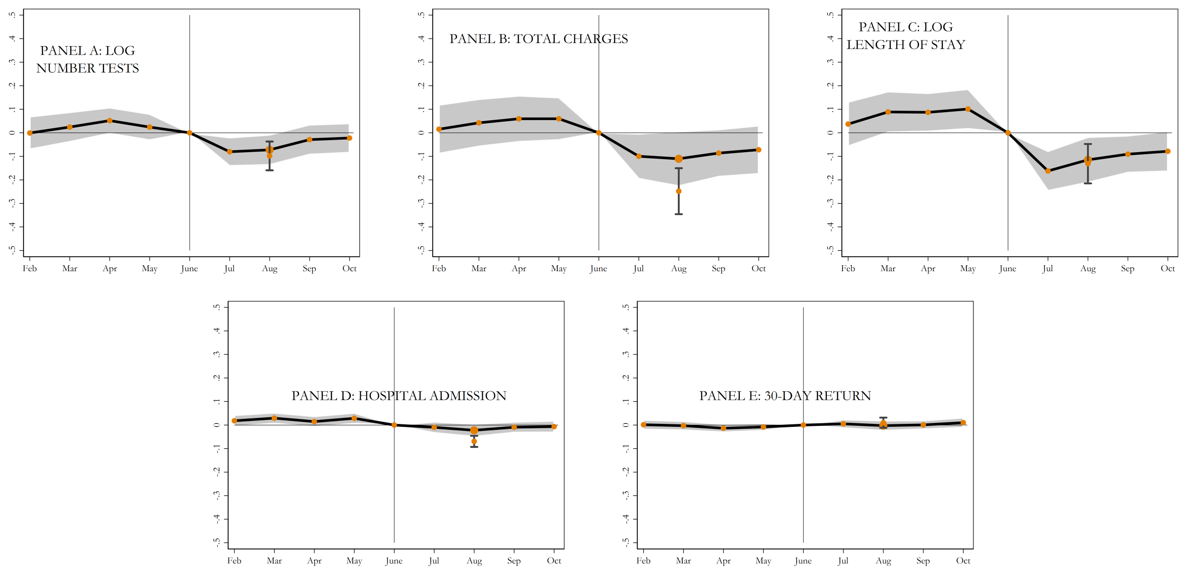
$$\begin{aligned} y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i. \end{aligned}$$

The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). In Columns 1-3, the dependent variable is a dummy if the patient is hospitalized. In Columns 4-6, the dependent variable is a dummy that takes the value of 1 if the patient returns to the ED within a 30-days period. All the controls are interacted with the year 2022 dummy. Panel B displays estimates of regressions of patients' evaluations in the ED on the period during which the technology was introduced, and interacted with the ward in which it was introduced. The estimating equation is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'\mathbf{X_i} + \epsilon_i$$

The main independent variable of interest is the interaction between being assigned to the private ward and arriving in the month of July or after. The model controls for insurance status (which subsumes the assigned ward), physician and month-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). For both panels where w indexes the ward to which the patient is assigned, t indexes the date/hour in which the patient arrived, and d indexes the physician to which the patient was assigned. The unit of observation is a case i arriving to the ED. Standard errors are clustered at the physician level.

FIGURE 1: THE IMPACT OF FASTER INFORMATION ACCESS: LEADS AND LAGS EVIDENCE



This Figure displays dynamic estimates of regressions of a case's length of stay in the ED on the period during which the technology was introduced and interacted with the ward in which it was introduced (i.e. private ward). The unit of observation is a case i arriving at the ED. This figure displays the coefficients β_j from estimating:

$$y_i = \sum_{t=2}^{10} \beta_j (Private_{w(i)} \times Monthj_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma' \mathbf{X_i} + \epsilon_\mathbf{i}$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived, and d indexes the physician to which the patient was assigned. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, and health markers upon admission). Standard errors are clustered at the physician level. To show longer dynamics we increase our sample from March to October (as the rest of the paper), to February to October.

Online Appendix

A.1 Mediation analysis

While identifying the quantitative importance of different mechanisms is a notoriously difficult exercise, in this subsection we examine whether any share in the decrease in the length of stay may be independent from the effects on the number of tests. Firstly, we estimate equation (2) while controlling for the number of tests ordered. We display the results in Column 4 Table A.5. We find there that the baseline triple-differences coefficient decreases in magnitude, and becomes not statistically significant. That is, the decrease in the length of stay becomes much smaller if one holds the number of tests ordered constant. We interpret this result as additional supportive evidence on the information acquisition mechanism—that increasing information speed had an effect in organizational outcomes through the laboratory tests that physicians chose to order.

While enlightening, the evidence in Column 4 Table A.5 can at best only be regarded as suggestive, given that the regressions are controlling for explicitly endogenous variables. For a more systematic analysis, we follow Heckman and Pinto (2015) in quantifying the relative importance of our mediating variable in the estimated decrease in length of stay. Heckman and Pinto (2015) consider an initial model $y_i = \beta_1 \cdot T_i + \beta_2 X_i + \epsilon_i$ where T_i is the introduction of the technology and X_i is a set of controls. The method decomposes the effect of the treatment into two parts:

$$\frac{dy}{dT} = \frac{\partial y}{\partial M}\frac{\partial M}{\partial T} + R \tag{A.1}$$

where M is the mediator –*Number of Tests*–. From (A.1) it is possible to isolate R given information on all other three elements. To do this, we substitute $\frac{dy}{dT}$ by the $\hat{\beta}$ from (2) where length of stay is the dependent variable. Secondly, we estimate $\hat{\beta}_{inter} = \frac{\partial M}{\partial T}$ from again regressing (2) but now having the mediator variable as the dependent variable. Lastly, we add the mediator M as an additional independent variable in (2) and obtain its estimated coefficient $\hat{\beta}_{med}$, which we take as an approximation to $\frac{\partial y}{\partial M}$. We can then define the ratio of mediator j as:

$$\frac{\hat{\beta}_{med(j)} \times \hat{\beta}_{inter(j)}}{\hat{\beta}}$$

We find that the mediator ratio is 48%, suggesting that around half of the treatment effect is due to physicians choosing to acquire different information (in this case, by ordering fewer tests). This implies that around 52% of the effect is independent of the mediating variable—which may be driven by other mechanisms such as the type of tests ordered, mechanical effects of being faster (22%), as well as the ability to learn better from the information acquired.

Figures and Tables

TABLE A.1: LENGTH OF STAY STATISTICS BY TIME (BEFORE AND AFTER) IN THE NON–PRIVATE WARD FROM 2018 TO 2022.

	Before						
	Non Private Ward						
	Mean	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**
Length of Stay (Days) – 2018	0.748	0.765	0.050	0.095	0.358	1.423	2.000
Length of Stay (Days) – 2019	0.767	0.725	0.072	0.158	0.427	1.335	2.000
Length of Stay (Days) – 2020	0.588	0.689	0.045	0.122	0.294	0.910	1.944
Length of Stay (Days) – 2021	0.792	0.788	0.029	0.079	0.424	1.704	2.000
Length of Stay (Days) – 2022	0.788	0.784	0.038	0.106	0.390	1.738	2.000
	After						
	Non Private Ward						
	Mean	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**
Length of Stay (Days) – 2018	0.764	0.761	0.053	0.097	0.388	1.442	2.000
Length of Stay (Days) – 2019	0.781	0.736	0.078	0.158	0.432	1.391	2.000
Length of Stay (Days) – 2020	0.770	0.743	0.036	0.117	0.429	1.379	2.000
Length of Stay (Days) – 2021	0.739	0.775	0.030	0.069	0.356	1.419	2.000
Length of Stay (Days) – 2022	0.949	0.774	0.089	0.230	0.750	2.000	2.000

Note: This table reports summary statistics for Length of Stay from 2018 to 2022. Statistics are presented for patients treated in the non-private ward, separately for the before (March–June) and after (July–October) periods used in the empirical analysis.

TABLE A.2: SUMMARY STATISTICS BY TIME AND WARD

	Before													
	Private							Non Private						
	Mean	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**	**Mean**	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**
Number of Tests	4.68	6.58	0.00	0.00	2.00	8.00	12.00	10.26	14.20	0.00	1.00	6.00	14.00	28.00
Total Charges	13.16	1.48	11.54	12.09	12.96	13.72	15.43	14.11	2.09	11.51	12.54	13.50	15.77	17.18
Length of Stay (Days)	0.40	0.55	0.03	0.07	0.17	0.39	1.27	0.79	0.78	0.04	0.11	0.39	1.74	2.00
Hospitalization	0.15	0.36	0.00	0.00	0.00	0.00	1.00	0.41	0.49	0.00	0.00	0.00	1.00	1.00
30-Day Return	0.12	0.33	0.00	0.00	0.00	0.00	1.00	0.13	0.34	0.00	0.00	0.00	0.00	1.00

	After													
	Private							Non Private						
	Mean	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**	**Mean**	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**
Number of Tests	4.49	6.34	0.00	0.00	1.00	7.00	12.00	11.68	13.64	0.00	1.00	8.00	16.00	29.00
Total Charges	13.13	1.49	11.53	12.03	12.92	13.69	15.46	14.59	2.09	11.88	12.90	14.52	16.21	17.48
Length of Stay (Days)	0.38	0.52	0.03	0.06	0.16	0.39	1.16	0.95	0.77	0.09	0.23	0.75	2.00	2.00
Hospitalization	0.14	0.35	0.00	0.00	0.00	0.00	1.00	0.52	0.50	0.00	0.00	1.00	1.00	1.00
30-Day Return	0.12	0.32	0.00	0.00	0.00	0.00	1.00	0.11	0.32	0.00	0.00	0.00	0.00	1.00

Note: This table displays summary statistics for the main outcomes. Length of stay is the time between triage and the departure of the patient from the ED (i.e. discharge or hospital admission). Number of tests is the number of laboratory tests ordered during the patient stay in the ED. Total charges (defined as the log of the amount in Million of Colombian Pesos (COP) charged by the hospital to the payer for each episode). The hospital admission dummy takes value one if the patient was admitted to the hospital instead of discharged home. The 30-day return dummy takes value one if the patient returns to the ED within 30 days. The sample includes between March and June 2022 (defined as before) and from July to October 2022 (defined as after).

TABLE A.3: SUMMARY STATISTICS BY TIME AND WARD—CONTROLS

	Before													
	Private							Non Private						
	Mean	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**	**Mean**	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**
Male Patient	0.38	0.49	0.00	0.00	0.00	1.00	1.00	0.42	0.49	0.00	0.00	0.00	1.00	1.00
Patient Age	46.67	19.05	22.00	31.00	44.00	60.00	74.00	47.26	19.72	24.00	30.00	43.00	63.00	76.00
Triage 1	0.01	0.07	0.00	0.00	0.00	0.00	0.00	0.10	0.30	0.00	0.00	0.00	0.00	1.00
Triage 2	0.04	0.21	0.00	0.00	0.00	0.00	0.00	0.64	0.48	0.00	0.00	1.00	1.00	1.00
Triage 3	0.63	0.48	0.00	0.00	1.00	1.00	1.00	0.23	0.42	0.00	0.00	0.00	0.00	1.00
Triage 4	0.31	0.46	0.00	0.00	0.00	1.00	1.00	0.02	0.15	0.00	0.00	0.00	0.00	0.00

	After													
	Private							Non Private						
	Mean	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**	**Mean**	**SD**	**p10**	**p25**	**p50**	**p75**	**p90**
Male Patient	0.39	0.49	0.00	0.00	0.00	1.00	1.00	0.49	0.50	0.00	0.00	0.00	1.00	1.00
Patient Age	47.18	19.05	23.00	31.00	45.00	61.00	75.00	50.53	20.16	24.00	32.00	50.00	67.00	78.00
Triage 1	0.01	0.08	0.00	0.00	0.00	0.00	0.00	0.13	0.34	0.00	0.00	0.00	0.00	1.00
Triage 2	0.05	0.21	0.00	0.00	0.00	0.00	0.00	0.66	0.47	0.00	0.00	1.00	1.00	1.00
Triage 3	0.69	0.46	0.00	0.00	1.00	1.00	1.00	0.19	0.39	0.00	0.00	0.00	0.00	1.00
Triage 4	0.25	0.43	0.00	0.00	0.00	0.00	1.00	0.02	0.14	0.00	0.00	0.00	0.00	0.00

Note: This table displays summary statistics for control variables. Male patient is a dummy equal to one when the patient is male and zero otherwise, the patient's age, and a dummy for each one of the four first triages. The sample includes between March and June 2022 (defined as after) and from July to October 2022 (defined as after).

TABLE A.4: CHANGES IN THE INFORMATION ACQUISITION PROCESS 99%, 97% AND 95%

	1% – 99%		**3% – 97%**		**5% — 95%**	
	Targeted	**Generic**	**Targeted**	**Generic**	**Targeted**	**Generic**
Post June X Private Ward X Year 2022	-0.042	-0.102	-0.035	-0.094	-0.034	-0.092
	(0.032)	(0.034)	(0.028)	(0.035)	(0.025)	(0.035)
Post June X Private Ward	Yes	Yes	Yes	Yes	Yes	Yes
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	62,650	62,650	62,683	62,683	62,683	62,683

Note: This table displays estimates of regressions of the logarithm of the number of tests on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy.

Columns 1 and 2 classify tests as targeted or generic using the 1% versus 99% thresholds of the distribution of laboratory test frequency. A test is defined as generic if its category appears in 50,000 or more episodes (corresponding to the 99% percentile of the distribution), and as targeted otherwise.

Columns 3 and 4 repeat the analysis using the 3% versus 97% thresholds, where the cutoff for being classified as generic is 16,000 episodes (the 97% percentile).

Columns 5 and 6 use the 5% versus 95% thresholds, with the cutoff set at 8,500 episodes (the 95% percentile).

The estimating equation is:

$$
\begin{aligned}
y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i.
\end{aligned}
$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived, and d indexes the doctor to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022.

The model controls for insurance status (which subsumes the assigned ward), doctor and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, and dummies for extreme vital sign markers upon admission) interacted with the 2022 dummy. Standard errors are clustered at the doctor level.

TABLE A.5: THE IMPACT OF FASTER INFORMATION ACCESS ON DECISION SPEED: LENGTH OF STAY

Dependent Variable = Log Length of Stay				
	(1)	**(2)**	**(3)**	**(4)**
	DiD	**Placebo DiD**	**Baseline DiDiD**	**Baseline DiDiD**
Year =	**2022**	**2019**	**2019&2022**	**2019&2022**
Post June X Private Ward	-0.247	-0.115	-0.115	-0.095
	(0.043)	(0.026)	(0.026)	(0.019)
Post June X Private Ward X Year 2022			-0.133	-0.051
			(0.051)	(0.033)
Log Number of Tests				0.777
				(0.011)
Patient Controls	Yes	Yes	Yes	Yes
Doctor Fixed Effects	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes
Interactions with 2022 dummy	No	No	Yes	Yes
Observations	29,108	32,492	61,600	61,600

Note: This Table displays estimates of regressions of a case's length of stay in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward). The unit of observation is a case i arriving at the ED. The estimating equation in Columns 1 and 2 is:

$$y_i = \beta(Private_{w(i)} \times Post_{t(i)}) + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \gamma'\mathbf{X_i} + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. Column 1 (2) shows the results of running this regression only for 2022 (2019). The main independent variable of interest is the interaction between being assigned to the private ward and arriving in the month of July or after. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). In Column 2 we repeat the Column 1 exercise on the placebo sample of 2019. In Columns 3 and, 4 the sample includes cases from both 2019 and 2022. The main independent variable of interest is the triple interaction between the private ward, the after-June dummy, and the 2022 dummy. The model controls for the interactions between all the controls and the year 2022 dummy. Standard errors are clustered at the physician level.

TABLE A.6: THE IMPACT OF FASTER INFORMATION ACCESS ON DECISION SPEED: LENGTH OF STAY BY TEST URGENCY

	Log. Length of Stay				
	Full	**Urgent More than 0**	**Non–Urgent Without Urgent**	**Urgent More than Mean %**	**Non–Urgent Less than Mean %**
Post June X Private Ward X 2022	-0.004	0.070	-0.008	0.106	-0.007
	(0.029)	(0.129)	(0.029)	(0.234)	(0.029)
Month Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Insurance Fixed Effects	Yes	Yes	Yes	Yes	Yes
Patient Controls	Yes	Yes	Yes	Yes	Yes
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes
Log. Number Tests	Yes	Yes	Yes	Yes	Yes
Observations	25,582	1,114	24,466	362	25,217

Note: This table displays estimates from regressions of a case's length of stay in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The unit of observation is a case i arriving at the ED.

The estimating equation is:

$$\begin{aligned} y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i. \end{aligned}$$

where w indexes the ward to which the patient is assigned, t indexes the month–year combination in which the patient arrived, and d indexes the doctor to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model includes month, year and insurance fixed effects, as well as log number of tests as control. Additionally, patient controls are included (age, gender, main diagnosis dummies, and indicators for extreme admission vital signs). Patient controls are interacted with the 2022 dummy. Standard errors are clustered at the doctor level.

Columns differ only in how the estimation sample is defined using laboratory test results, which include both the test outcome and reference ranges. Column 1 (Full sample with test information) includes all episodes with at least one test for which both the result and the corresponding reference range are available, so that test abnormality can be defined. For columns 2 and 3, an episode is classified as urgent if it has at least one test result outside its reference range, and as non–urgent otherwise. Columns 3 and 4 use an intensity–based urgency definition based on the share of tests outside the reference range within an episode: for each episode, we compute the fraction of tests with results outside the reference range; we then compute the mean of this fraction in the full test–information sample (Column 1) and classify episodes with a fraction above the mean as urgent and those at or below the mean as non–urgent.

TABLE A.7: THE IMPACT OF FASTER INFORMATION ACCESS ON DECISION SPEED: LENGTH OF STAY WITH AND WITHOUT TESTS

	(1) All Episodes	(2) Without Tests	(3) With Tests
Post June X Private Ward X Year 2022	-0.133	-0.072	-0.156
	(0.051)	(0.067)	(0.052)
Post June X Private Ward	Yes	Yes	Yes
Patient Controls X Year 2022	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes
Post June X Year 2022	No	No	No
Private Ward X Year 2022	No	No	No
Observations	61,600	20,892	37,270

Note: This table displays estimates of regressions of a case's length of stay in the ED on the period during which the informational dashboard was introduced (i.e. post 18 June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The unit of observation is a case i arriving at the ED. The estimating equation is:

$$\begin{aligned} y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i. \end{aligned}$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. Column 1 includes all episodes. Column 2 only includes episodes in which no tests were requested, while Column 3 only includes episodes in which there was at least one requested test. Standard errors are clustered at the physician level.

TABLE A.8: THE IMPACT OF FASTER INFORMATION ACCESS ON DECISION SPEED: CASES REQUIRING TESTS

Dependent Variable =	(1) Log. Number of Tests	(2) Total Charges	(3) Log. Length of Stay	(4) Hospital.	(5) 30–Day Return
Post June X Diag. with Test	-0.197	-0.173	-0.203	-0.040	0.000
	(0.036)	(0.062)	(0.050)	(0.017)	(0.017)
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes
Dr. Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes
Ins. Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	36,100	36,066	35,712	36,100	36,100

Note:This table displays estimates of regressions of a case's different medical outcomes in the ED on the period during which the technology was introduced (i.e. after June), interacted with a dummy equal to 1 for diagnosis with tests (defined as a diagnosis that received at least one test in the first semester of 2022 (excluding June)) and 0 otherwise. In this table, we focus only on private episodes in 2022. The estimating equation is:

$$y_i = \beta(Post_{t(i)} \times DiagTest_i) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma'(\mathbf{X_i}) + \epsilon_i$$

The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). In column 1, the dependent variable is the case's length of stay in the ED. In column 2, the dependent variable is the logarithm of the number of tests. In column 3, the dependent variable is the total charge (defined as the log of the amount in Colombian Pesos (COP) charged by the hospital to the payer for each episode). In Column 4, the dependent variable is a dummy if the patient is hospitalized. The last column, the dependent variable is a dummy that takes the value of 1 if the patient returns to the ED within a 30–day period. Standard errors are clustered at the physician level.

TABLE A.9: THE IMPACT OF FASTER INFORMATION ACCESS ON DECISION SPEED: LENGTH OF STAY BY WHETHER TESTS WERE HISTORICALLY REQUIRED

Dependent Variable =	**Log. Length of Stay**		
	Full Sample	**ICD with Tests**	**ICD without Tests**
Post June X Private Ward X 2022	-0.123	-0.139	-0.059
	(0.028)	(0.050)	(0.038)
Coef. Diffs. P–value	–	.007	
Patient Controls X Year 2022	Yes	Yes	Yes
Dr. Fixed Effects X Year 2022	Yes	Yes	Yes
Ins. Fixed Effects X Year 2022	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes
Observations	61,612	26,924	31,162

This Table displays estimates of regressions of a case's length of stay in the ED on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. The unit of observation is a case i arriving at the ED. The estimating equation is:

$$\begin{aligned} y_i = &\beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\ &+ \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\ &+ \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i. \end{aligned}$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission). The dependent variables are the log length of stay, total charge (defined as the log of the amount in Colombian Pesos (COP) charged by the hospital to the payer for each episode), a dummy if the patient is hospitalized, a dummy that takes the value of 1 if the patient returns to the ED within a 30-days period and log number of tests. All the controls are interacted with the year 2022 dummy. Results are presented for the full sample, as well as sample cuts based on the median doctor age in 2019. Certain conditions, such as fractures, generally do not require laboratory tests, while others, like infections or metabolic disorders, often do. To differentiate these cases, ICD-10 groups were classified based on their typical need for tests. Conditions requiring tests include infectious diseases (A00-B99), tumors (C00-D48), blood and immune disorders (D50-D89), endocrine and metabolic diseases (E00-E90), nervous system disorders (G00-G99), cardiovascular diseases (I00-I99), respiratory diseases (J00-J99), digestive diseases (K00-K93), genitourinary diseases (N00-N99), pregnancy and perinatal conditions (O00-O99, P00-P96), and congenital malformations (Q00-Q99), as these often require blood tests, imaging, or biopsies for diagnosis. Conversely, conditions that generally do not require tests include mental and behavioral disorders (F00-F99), eye diseases (H00-H59), ear and mastoid disorders (H60-H95), skin and subcutaneous tissue diseases (L00-L99), musculoskeletal disorders (M00-M99), general symptoms and abnormal clinical findings (R00-R99), injuries and external causes (S00-T98, V00-Y99), and health-related factors and contact with health services (Z00-Z99), which are primarily diagnosed through clinical evaluation. All ICD-10 codes starting with 'U' (U00-U99) have been classified as requiring diagnostic tests. While some codes in this category are used for statistical reporting or administrative purposes, the majority represent emerging infectious diseases, antimicrobial resistance, or other conditions that typically require laboratory confirmation or imaging. To maintain consistency, all 'U' codes have been grouped under conditions that require testing.

TABLE A.10: THE IMPACT OF FASTER INFORMATION ACCESS ON DECISION SPEED: EVIDENCE ON COGNITIVE MECHANISMS

PANEL A: BUSY AND NON–BUSY HOURS ANALYSIS

	Log. Number of Tests		Total Charges		Log. Length of Stay		Hospitalization		30–Day Return	
	Busy	Non-Busy	Busy	Non-Busy	Busy	Non-Busy	Busy	Non-Busy	Busy	Non-Busy
Post June X Private Ward X Year 2022	-0.438	-0.328	-0.497	-0.108	-0.533	-0.132	-0.144	-0.057	0.053	0.001
	(0.242)	(0.077)	(0.154)	(0.043)	(0.211)	(0.062)	(0.051)	(0.017)	(0.063)	(0.015)
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,151	43,773	18,190	43,839	17,889	43,064	18,190	43,839	18,190	43,839

PANEL B: COMMON AND UNCOMMON DIAGNOSIS. 50–50 ANALYSIS

	Log. Number of Tests		Total Charges		Log. Length of Stay		Hospitalization		30–Day Return	
	Uncommon	Common	Uncommon	Common	Uncommon	Common	Uncommon	Common	Uncommon	Common
Post June X Private Ward X Year 2022	-0.194	-0.101	-0.407	-0.222	-0.427	-0.130	-0.129	-0.057	0.026	-0.043
	(0.065)	(0.067)	(0.091)	(0.096)	(0.092)	(0.082)	(0.023)	(0.022)	(0.017)	(0.023)
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	34,298	24,930	34,219	24,897	33,735	24,385	34,298	24,930	34,298	24,930

PANEL C: COMMON AND UNCOMMON DIAGNOSIS. 80–20 ANALYSIS

	Log. Number of Tests		Total Charges		Log. Length of Stay		Hospitalization		30–Day Return	
	Uncommon	Common	Uncommon	Common	Uncommon	Common	Uncommon	Common	Uncommon	Common
Post June X Private Ward X Year 2022	-0.176	0.017	-0.405	-0.003	-0.378	0.019	-0.127	0.005	0.024	-0.058
	(0.054)	(0.107)	(0.078)	(0.142)	(0.079)	(0.130)	(0.020)	(0.030)	(0.014)	(0.045)
Patient Controls X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects X Year 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	48,569	10,109	48,486	10,088	47,651	9,977	48,569	10,109	48,569	10,109

Note: This table displays estimates from regressions of a case's log length of stay, log of the number of tests, total charges, hospitalization, and 30-day return in the ED on the period during which the informational dashboard was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. All panels report results from the same empirical specification, use the same set of dependent variables, and focus on the same main independent variable; panels differ only in the definition of the estimation sample. The unit of observation is a case i arriving at the ED.

The estimating equation is:

$$
\begin{aligned}
y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i .
\end{aligned}
$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date–hour combination) in which the patient arrived, and d indexes the physician to whom the patient is assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving after the dashboard introduction, and arriving in 2022. The model includes insurance status fixed effects (which subsume ward assignment), physician fixed effects, and hour fixed effects, as well as patient controls for age, gender, main diagnosis indicators, and indicators for extreme vital signs upon admission; all controls and fixed effects are interacted with the year-2022 dummy. Standard errors are clustered at the physician level.

Panel A reports results separately for busy and non–busy hours. Busy hours are defined as those in which the patient–physician ratio exceeds the 75th percentile of its distribution, which is close to two patients per physician.

Panel B reports results separately for uncommon and common diagnosis groups defined according to their share of total tests. Diagnosis groups are ordered by their frequency of tests, and the most frequent groups are classified as "common" until they cumulatively account for 50% of all tests; the remaining groups are classified as "uncommon." Under this definition, three diagnosis groups together account for approximately 50% of all tests. Labels are constructed following this cumulative–frequency logic.

Panel C applies the same cumulative–frequency logic but defines common diagnosis groups as those that together account for 80% of all tests, with the remaining groups classified as uncommon. Thus, the distinction between common and uncommon diagnoses reflects the cumulative share of tests explained by each group rather than the number of diagnosis categories.

TABLE A.11: LACK OF EVIDENCE ON PEER EFFECTS

	Log. Number of Tests		Total Charges		Log. Length of Stay		Hospitalization		30-Day Return	
	Number of Drs.		**Number of Drs.**		**Number of Drs.**		**Number of Drs.**		**Number of Drs.**	
	1	**2+**	**1**	**2+**	**1**	**2+**	**1**	**2+**	**1**	**2+**
Post June X Private Ward X 2022	-3.617	-0.294	-0.605	-0.055	-0.251	-0.068	-0.174	-0.020	0.027	-0.018
	(0.294)	(1.306)	(0.055)	(0.131)	(0.021)	(0.056)	(0.014)	(0.040)	(0.018)	(0.036)
Patient Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Doctor Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurance Status Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	46,629	12,829	46,562	12,796	46,629	12,829	46,629	12,829	46,629	12,829

Note: This Table displays estimates of regressions of a case's number of physicians working in an episode on patient outcomes on the period during which the informational dashboard was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) and with the year 2022 dummy. To define the number of physicians working in an hour–ward–episode, we count the total number of physicians that are directly treating patients, that are requesting tests for patients, or that close any ED episode. The estimating equation is:

$$
\begin{aligned}
y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i.
\end{aligned}
$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date–hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. Sample changes for each column depending on the number of physicians working on an episode (treatment physician and other physicians who request tests). Standard errors are clustered at the physician level.

TABLE A.12: LENGTH OF TEST PROCESSING

	(1)	(2)	(3)	(4)	(5)
Post June X Private Ward X Year 2022	-0.001	0.005	0.009	0.045	-0.026
	(0.035)	(0.034)	(0.034)	(0.033)	(0.031)
Patient Controls X Year 2022	Yes	No	Yes	Yes	Yes
Doctor Fixed Effects	Yes	Yes	No	Yes	Yes
Insurance Status Fixed Effects	Yes	Yes	Yes	No	Yes
Date X Hour Fixed Effects	Yes	Yes	Yes	Yes	No
Observations	37,555	37,555	37,701	37,555	39,507

Note: This Table displays estimates of regressions of a case's the total time that the test took to be delivered in the laboratory department on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward) interacted with the year 2022. The unit of observation is a case i arriving at the ED. The estimating equation is:

$$\begin{aligned} y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\ & + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i. \end{aligned}$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date–hour of day combination) in which the patient arrived, and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour–fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the 2022 dummy. Standard errors are clustered at the physician level.

TABLE A.13: EFFECT OF THE DASHBOARD ON LENGTH OF STAY FOR URGENT CASES

Dependent Variable = Log. Length of Stay	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**
Urgent X Post	0.120	0.099	0.135	0.099	0.152
	(0.122)	(0.123)	(0.057)	(0.056)	(0.119)
Date X Hour FE	Yes	Yes	No	No	Yes
Doctor FE	No	No	Yes	Yes	Yes
Patient Controls	No	Yes	No	Yes	Yes
Observations	954	952	2,435	2,434	923

Note: This table reports estimates from regressions examining whether the introduction of the screen affected patient Length of Stay for emergency department episodes involving urgent laboratory test results. The construction of the estimation sample proceeds in several steps.

First, laboratory test data are used to identify urgent test results at the episode level. For each laboratory observation, a test result is classified as urgent if its value lies outside the clinically urgent (or panic) range defined by the laboratory reference thresholds. An episode is classified as urgent if it includes at least one such urgent test result. The episode-level urgent indicator equals one if any test within the episode is urgent, and zero otherwise.

Second, the analysis sample is restricted to episodes occurring in the private ward, with non-missing and positive Length of Stay, and to episodes taking place in calendar year 2022. The dependent variable y_i is the logarithm of patient Length of Stay, measured as the time elapsed between admission and exit from the emergency department.

A post-treatment indicator, $Post(i)$, equals one for episodes occurring on or after June 28, 2022, corresponding to the introduction of the screen, and zero otherwise. The main explanatory variable is the interaction $Urgent(i) \times Post(i)$, which captures whether episodes involving urgent test results experienced differential changes in Length of Stay after the screen was introduced. We estimate the following equation:

$$y_i = \beta\,[Urgent(i) \times Post(i)] + \mathbf{X}_i\delta + \alpha_{d(i)} + \pi_{t(i)} + \varepsilon_i,$$

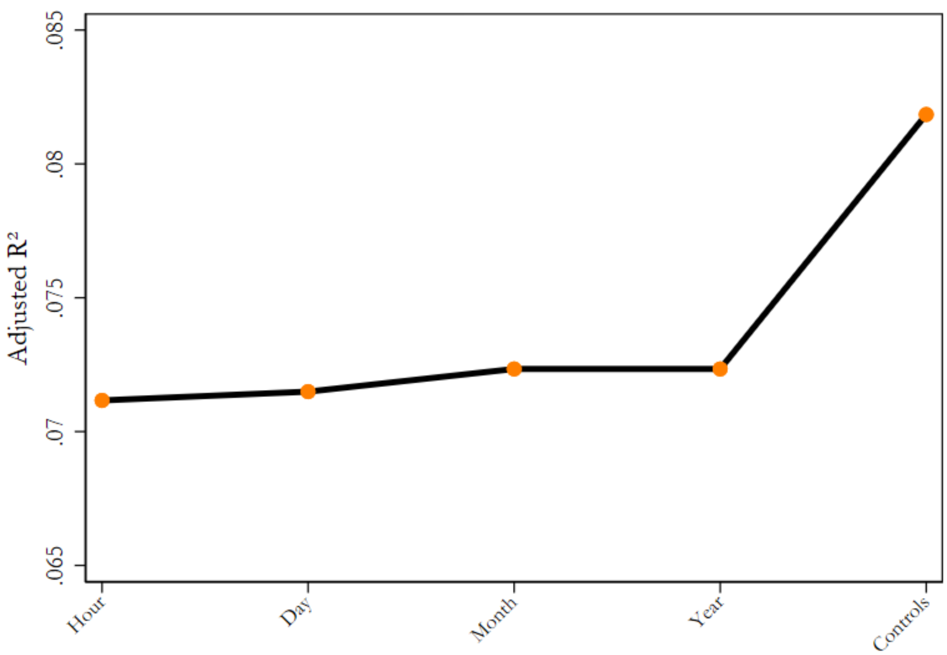
where $\mathbf{X}_i$ denotes patient-level controls, $\alpha_{d(i)}$ are doctor fixed effects, and $\pi_{t(i)}$ are date-by-hour fixed effects.

Each column in the table corresponds to a different specification. Column 1 includes date-by-hour fixed effects only. Column 2 adds patient-level controls to this specification. Column 3 includes doctor fixed effects only. Column 4 adds patient-level controls to the doctor fixed effects specification. Column 5 includes date-by-hour fixed effects, doctor fixed effects, and patient-level controls simultaneously. The unit of observation in all regressions is an episode. Standard errors are clustered at the doctor level.

FIGURE A.1: Dashboard and Environment

Note: This figure displays a screenshot of the dashboard. An *Orden* corresponds to a set of required tests, and an episode may include multiple *ordenes*. The dates and times shown beneath each circle were added later and were not visible during the intervention period. Test status is indicated by color-coded circles: white (pending), blue (validating), green (completed and normal), yellow (concerning), and red (critical). During the intervention, the dashboard also displayed the patient's national ID number.

FIGURE A.2: INCREMENTAL EXPLANATORY POWER OF OBSERVABLES IN TEST PROCESSING TIME



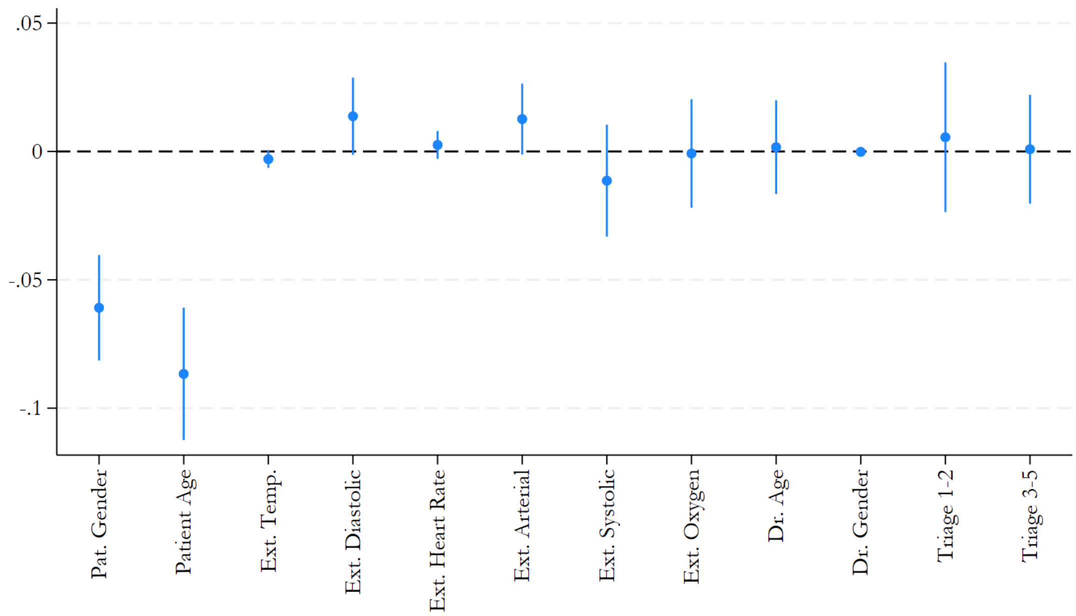
Note: This table presents the adjusted R^2 values for regressions analyzing how observable variables influence the length of test processing. The key explanatory variable is the interaction between the period in which the technology was introduced (i.e., after June), the ward where it was implemented (i.e., private ward), and a dummy for the year 2022. The unit of observation is a case i arriving at the ED, and the estimation equation for the most comprehensive specification (Controls) includes both patient-level observables (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital sign markers upon admission) and time-related observables associated with the patient's admission date. The estimation equation is:

$$
\begin{aligned}
y_i = \; & \beta_3 \big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)} \big) \\
& + \beta_2^{(1)} \big(Private_{w(i)} \times Post_{t(i)} \big) + \beta_2^{(2)} \big(Private_{w(i)} \times 2022_{t(i)} \big) + \beta_2^{(3)} \big(Post_{t(i)} \times 2022_{t(i)} \big) \\
& + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}' \big(\mathbf{X}_i \times 2022_{t(i)} \big) + \epsilon_i .
\end{aligned}
$$

where j indexes the hour, w the day, d the month, and t the year to which the patient is assigned. The table presents results from a stepwise inclusion of fixed effects:

- Hour: Includes only hour fixed effects, without patient controls.
- Day: Adds day fixed effects.
- Month: Includes month fixed effects in addition to hour and day fixed effects.
- Year: Incorporates all fixed effects (Hour, Day, Month, and Year) but excludes patient controls.
- Controls: Includes all fixed effects and patient controls, allowing us to assess the contribution of observable patient

FIGURE A.3: STABILITY OF PATIENT COMPOSITION AROUND TECHNOLOGY INTRODUCTION



characteristics to the variation in test processing time.

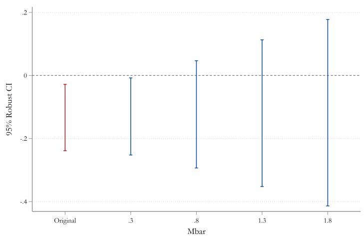
The adjusted R^2 measures the proportion of variance in test processing time explained by the model, penalizing for the inclusion of additional regressors to ensure a more accurate assessment of explanatory power. Comparing specifications helps to quantify the incremental explanatory power of including patient characteristics and time fixed effects. Higher adjusted R^2 values in more comprehensive specifications indicate that observable factors play a meaningful role in explaining differences in test processing time, beyond random variation.

Note: This figure displays coefficients of different regressions for the following dependent variables y_i: Patient Gender (dummy defined as 1 for men and 0 for women), Patient Age dummy defined as 1 for patients whose age is above the median and 0 otherwise, Extreme Vital Signs (diastolic pressure, heart rate, arterial pressure, systolic pressure and oxygen saturation), Dr. Age dummy defined as 1 for physicians whose age is above the median and 0 otherwise, Dr. Gender (dummy defined as 1 for men and 0 for women), low triage dummy defined as 1 for triages 1 and 2 and 0 otherwise. The estimating equation is:

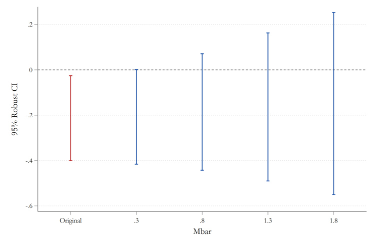
$$\begin{aligned}
y_i &= \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\
&\quad + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\
&\quad + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i.
\end{aligned}$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date–hour of day combination) in which the patient arrived and d indexes the physician to which the patient was assigned. The model controls for insurance status (which subsumes the assigned ward), physician and hour fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, as well as dummies for extreme vital signs markers upon admission) interacted with the year 2022 dummy. When the variable is the dependent variable, it is excluded from the control list. For instance, for the first variable, the dependent variable is patient gender and in this case, is excluded from the list $\mathbf{X_i}$. Standard errors are clustered at the physician level.

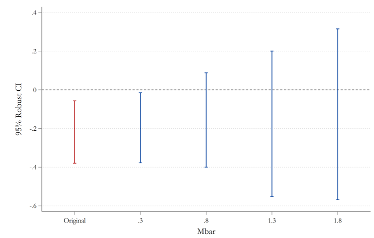
FIGURE A.4: HONEST DiD



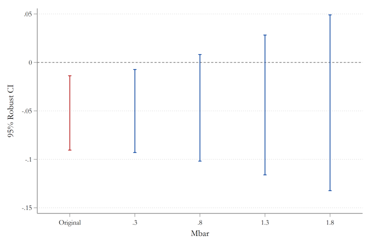
(a) Number of Tests



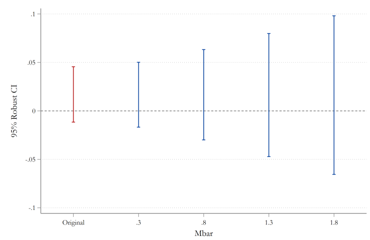
(b) Total Charges



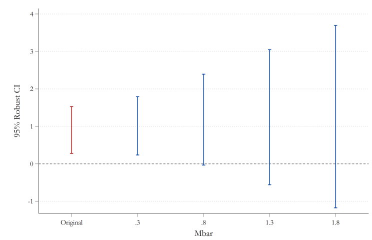
(c) Length of Stay



(d) Hospitalization



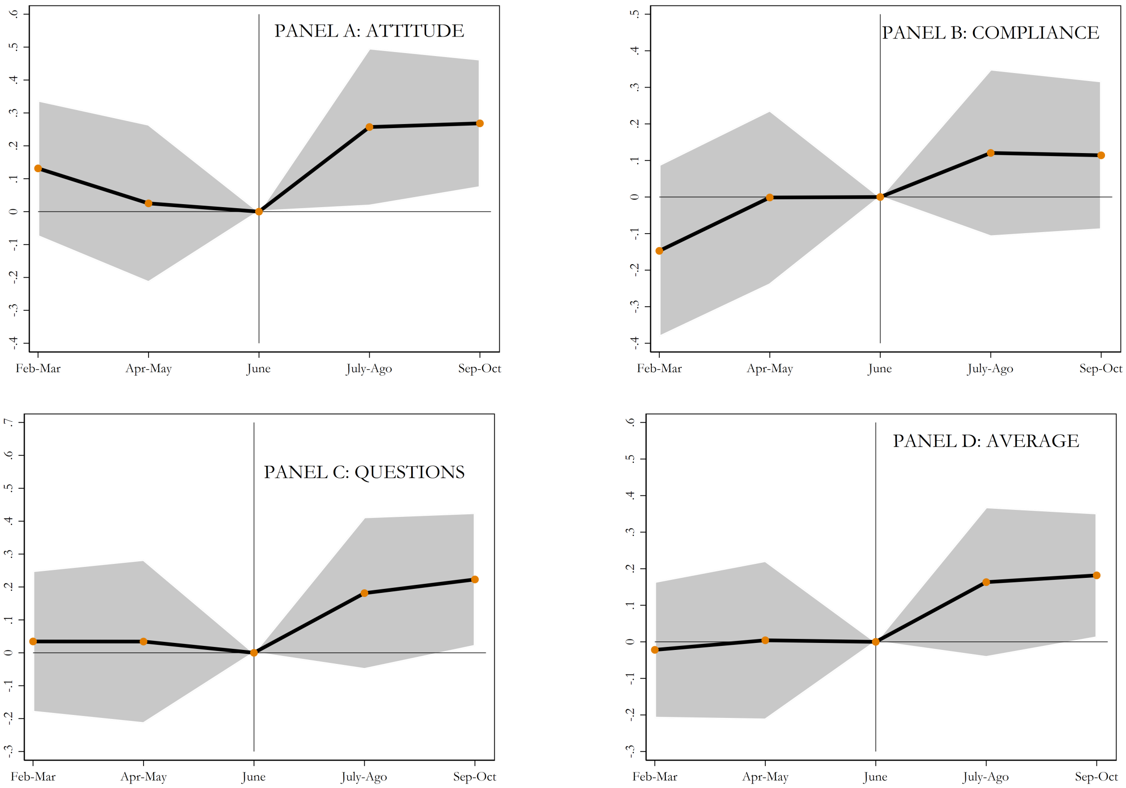
(e) 30–Day Return



(f) Average Patient Satisfaction

Note: These figures display coefficients for sensitivity analysis before and after the treatment. It shows a robust confidence interval for different values of proportional violation of parallel trends assumption. The value equal to 1, for instance, imposes that the post–treatment violation of parallel trends is no longer than the worst pre–treatment violation of parallel trends (between consecutive periods). Likewise, a value of 2 implies that the post–treatment violation of parallel trends is no more than twice that in the pre–treatment period. Where the coefficients cross the zero line (breakdown value), it means that the significant result is robust to allowing for violations of parallel trends up to the value and as big as the max violation in the pre–treatment period.

FIGURE A.5: LEADS AND LAGS EVIDENCE DOUBLE-DIF

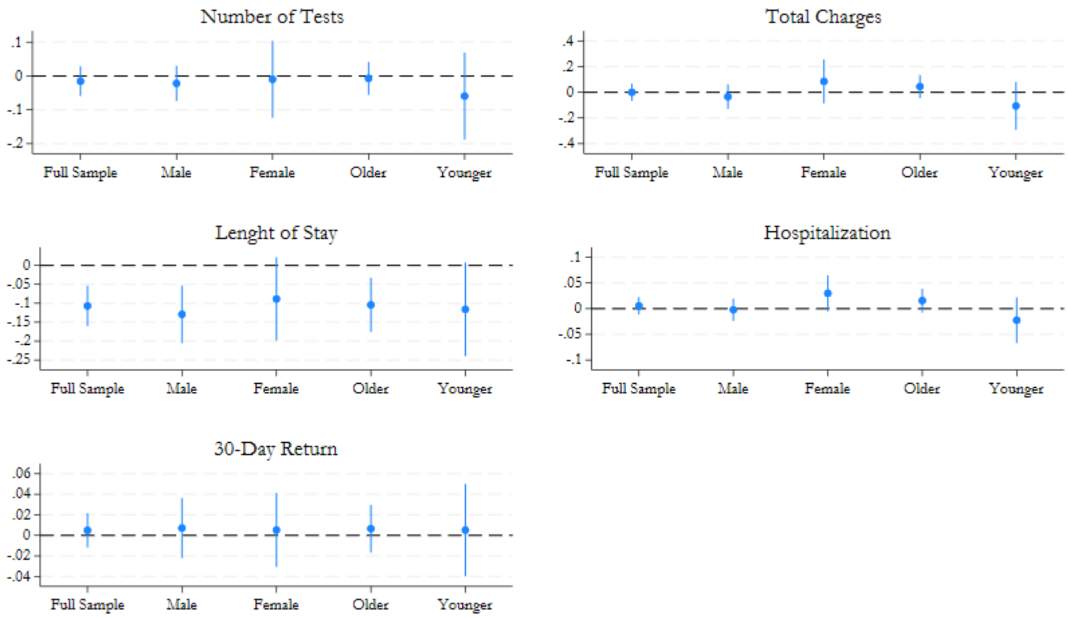


This Figure displays dynamic estimates of regressions of a case's survey evaluation on the period during which the technology was introduced (i.e. after June), interacted with the ward in which it was introduced (i.e. private ward). The unit of observation is a case i arriving at the ED. This figure displays the coefficients β_t from estimating:

$$y_i = \sum_{t=2-3,4-5}^{7-8,9-10} \beta_t (Private_{w(i)} \times Month_{t(i)}) + \alpha_{d(i)} + \theta_{w(i)} + \pi_{t(i)} + \gamma' \mathbf{X_i} + \epsilon_i$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e. date/hour of day combination) in which the patient arrived, and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward and arriving after June. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, and health markers upon admission). Standard errors are clustered at the physician level.

FIGURE A.6: HETEROGENEITY BY DOCTOR GENDER AND AGE



Note: This figure presents coefficient estimates from regressions of a case's different medical outcomes in the ED on the period during which the technology was introduced (i.e. after June), interacted with a dummy equal to 1 for diagnosis without tests en the pre–period (before June 2022) and 0 otherwise. The estimating equation is:

$$
\begin{aligned}
y_i = {} & \beta_3\big(Private_{w(i)} \times Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \beta_2^{(1)}\big(Private_{w(i)} \times Post_{t(i)}\big) + \beta_2^{(2)}\big(Private_{w(i)} \times 2022_{t(i)}\big) + \beta_2^{(3)}\big(Post_{t(i)} \times 2022_{t(i)}\big) \\
& + \alpha_{d(i)} + \theta_{in(i)} + \pi_{t(i)} + \boldsymbol{\gamma}'\big(\mathbf{X}_i \times 2022_{t(i)}\big) + \epsilon_i.
\end{aligned}
$$

where w indexes the ward to which the patient is assigned, t indexes the exact hour (i.e., date/hour of day combination) in which the patient arrived, and d indexes the physician to which the patient was assigned. The main independent variable of interest is the interaction between being assigned to the private ward, arriving in the months between July and October, and arriving in 2022. The model controls for insurance status (which subsumes the assigned ward), physician and hour-fixed effects, as well as patient controls (age, gender, dummies for the main diagnosis, and dummies for extreme vital signs markers upon admission). All the patient controls are interacted with the year 2022 dummy. Results are presented for the full sample, as well as for sample splits based on doctor gender and age (young and old doctors were classified using age data from the years 2019 and 2022. For each doctor, the average of their recorded age across these two years was calculated. Based on this average, the 50th percentile (median) was computed, and doctors whose average age was above the median were classified as "Old", while those below the median were classified as "Young").

A.2 Measurement and Variable Definitions

This section describes the construction and interpretation of the main dependent and independent variables used in the analysis. The goal is to clarify how each theoretical construct is operationalized in the empirical setting.

A.2.1 Dependent Variables

Number of Laboratory Tests The number of tests is defined as the total count of laboratory tests ordered during a patient's emergency department (ED) episode. Tests include all in-house laboratory diagnostics (e.g., blood panels, urinalyses) ordered between admission and exit from the ED. This variable captures physicians' information acquisition choices that inform their decisions. In the main analyses, we use the logarithm of the number of tests to account for right skewness. This outcome is used to test Hypothesis 1 on whether faster information access leads to less but more targeted information acquisition.

Composition and Timing of Laboratory Tests

We also construct outcomes that capture the composition and timing of information acquisition. We classify tests as generic or targeted based on how frequently each test category is used across all emergency department episodes. A test is defined as generic if its category appears in at least 50,000 episodes (the 99th percentile of the test-frequency distribution), and as targeted otherwise. We assess robustness using alternative frequency cutoffs corresponding to the 97th percentile (16,000 episodes) and the 95th percentile (8,500 episodes).

To examine when information acquisition responds to faster access, we also study heterogeneity across the episode's length of stay. We split episodes into quartiles based on total duration and estimate effects separately for each quartile, allowing us to distinguish responses early versus later in the clinical decision process.

Total Charges Total charges measure the total amount billed by the hospital to the payer for each ED episode. Charges include laboratory tests, medications, consumables, and physician fees incurred between admission and discharge or hospital admission. The variable is measured in Colombian Pesos and log-transformed in the main specifications. This outcome captures the cost implications of information acquisition and decision-making efficiency.

Length of Stay Length of stay (LOS) is defined as the time elapsed between a patient's admission to the ED and their exit, either through discharge home or admission to the hospital. LOS is measured in continuous time and log-transformed in the main analyses. For non-critical patients—the focus of this study—exit from the ED is primarily determined by the physician's disposition decision. Accordingly, LOS serves as a proxy for the time required to reach a clinical decision, consistent with prior work in emergency department settings. This outcome is used to test Hypothesis 2 on whether faster information access leads to faster decisions.

Hospitalization Hospitalization is a binary indicator equal to one if the patient is admitted to the hospital following the ED episode and zero if the patient is discharged home. This variable captures an important dimension of decision outcomes and quality of care, particularly in non-critical cases where admission typically reflects unresolved uncertainty or lack of sufficient improvement during the ED stay. This outcome is used to test Hypothesis 3 on whether faster information access leads to improved decisions.

Thirty-Day Return The 30-day return variable is a binary indicator equal to one if the patient returns to the ED within 30 days of discharge and zero otherwise. This outcome serves as a standard measure of downstream quality and safety, capturing whether faster or more efficient decisions lead to premature discharge or deterioration after leaving the ED. This outcome is used to test Hypothesis 3 on whether faster information access leads to improved decisions.

Patient Satisfaction Patient satisfaction is measured using post-visit surveys administered to a randomly selected subset of patients in 2022. We focus on three survey items: (i) perceived attitude of the medical staff, (ii) perceived compliance with good medical practices, and (iii) perceived quality of information provision and responsiveness to questions. Each outcome is analyzed separately, and we also consider their average. Responses are coded on a standardized numerical scale. Due to data availability, analyses using these outcomes rely on a difference-in-differences specification rather than the triple-differences design used for administrative outcomes. This outcome is used to test Hypothesis 3 on whether faster information access leads to improved decisions.

A.2.2 Independent Variables

Treatment Indicator The main treatment variable captures increased information access speed induced by the introduction of the dashboard. It is defined as the interaction between assignment to the private insurance ward, arrival after the dashboard introduction (post-June), and arrival in year 2022. This triple interaction isolates the effect of faster information access by comparing outcomes across wards, time periods, and years.

Post Indicator The post indicator equals one for episodes occurring after the introduction of the dashboard (July onward) and zero otherwise. In the main specifications, this indicator is interacted with ward assignment and year to account for seasonal patterns and pre-existing differences across wards.

Ward Assignment Ward assignment is determined by patients' insurance status and triage level and is not chosen by physicians or patients. Patients assigned to the private insurance ward constitute the treated group in 2022, while patients in the regular ward serve as the control group throughout the sample period.

Control Variables All regressions include a rich set of pre-determined controls measured at admission: patient age, gender, triage category, main diagnosis indicators, and indicators for extreme vital signs. We also include physician fixed effects, insurance fixed effects (which subsume ward assignment), and detailed date-by-hour fixed effects. In triple-differences specifications, patient controls and fixed effects are interacted with the year-2022 indicator to allow for differential evolution over time.

Together, these variables allow us to map the theoretical constructs of information acquisition, decision speed, and decision quality to observable outcomes while isolating the effect of faster information access from confounding operational or compositional changes.